PHYSICS

1) Given below are two statements :
Statement-I: The magnetic force acting on a moving charge can never change speed of the charge.
Statement-II: An electric field can only speed up or slow down a moving charge but can never deflect the charge.
In the light of the above statements. choose the most appropriate answer form the options given below.

(1) Both statement I and statement II are incorrect

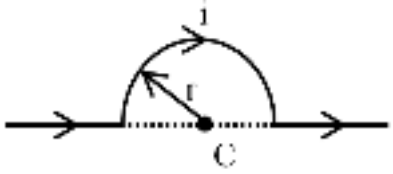
(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct

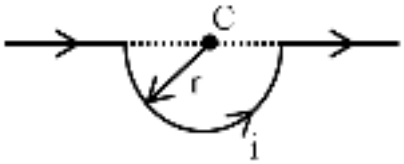
(4) Both statement I and statement II are correct

2) In which of the following cases, magnetic field vector at the centre C is found to be same ?

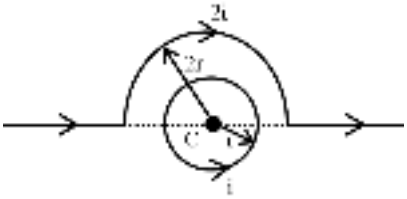
(a)



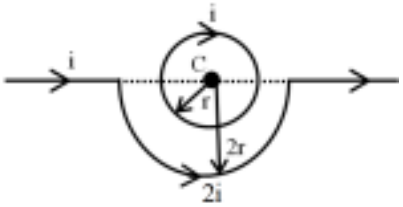
(b)



(c)



(d)



(1) a and b

(2) a and c

(3) b and c

(4) a, c and d

3) A closely wound solenoid 80 cm long has 5 layers of windings of 400 turns each. The diameter of the solenoid is 1.8 cm. If the current carried is 8.0 A. estimate the magnitude of B inside the solenoid

near its centre.

(1) zero

(2) $8\pi \times 10^{-3}$ T

(3) $15\pi \times 10^{-3}$ T

(4) $\pi \times 10^{-3}$ T

4) Two long, thin wires distant 'a' apart exert a force F on one another when current through each wire is 'i'. the distance between the wire is doubled and the current is decreased to i/4. The force they exert on one another now :

(1) $\frac{F}{8}$

(2) $\frac{F}{16}$

(3) $\frac{2F}{4}$

(4) $\frac{F}{32}$

5) A long straight wire along the z-axis carries a current *i* in the negative z-direction. The magnetic field vector $\vec{B}$ at a point having co-ordinates (x, y) in the z = 0 plane is :-

(1) $\frac{\mu_0 i(y\hat{i} - x\hat{j})}{2\pi(x^2 + y^2)}$

(2) $\frac{\mu_0 i(y\hat{i} + x\hat{j})}{2\pi(x^2 + y^2)}$

(3) $\frac{\mu_0 i(y\hat{j} - x\hat{i})}{2\pi(x^2 + y^2)}$

(4) $\frac{\mu_0 i(y\hat{j} + x\hat{i})}{2\pi(x^2 + y^2)}$

6) A series LCR circuit has L = 0.01 H, R = 10 Ω and C = 1 μF and it is connected to ac voltage of amplitude (V_m) 50 V. At frequency 60% lower than resonant frequency, the amplitude of current will be approximately :

(1) 466 mA

(2) 312 mA

(3) 238 mA

(4) 196 mA

7) In a series LR circuit, power of 400 W is dissipated from a source of 250 V, 50 Hz. The power factor of the circuit is 0.8. In order to bring the power factor to unity, a capacitor of value C is added in series to the L and R. Taking the value of C as $\left(\frac{n}{3\pi}\right)\mu F$, then value of n is _____ .

(1) 200

(2) 100

(3) 400

(4) 50

8) An EM wave from air enters a medium. The electric fields are $\vec{E}_1 = E_{01}\hat{x}\cos\left[2\pi\nu\left(\frac{z}{c}-t\right)\right]$ in air and $\vec{E}_2 = E_{02}\hat{x}\cos[k(2z-ct)]$ in medium, where the propagation constant k and frequency v refer to their values in air. The medium is non-magnetic. If $\in_{r_1}$ and $\in_{r_2}$ refer to relative permittivity of air and medium respectively, which of the following options is correct ?

(1) $\frac{\in_{r_1}}{\in_{r_2}} = 2$

(2) $\frac{\in_{r_1}}{\in_{r_2}} = \frac{1}{4}$

(3) $\frac{\in_{r_1}}{\in_{r_2}} = \frac{1}{2}$

(4) $\frac{\in_{r_1}}{\in_{r_2}} = 4$

9) A circular coil of radius 2.00 cm has 50 turns. A uniform magnetic field B = 0.200 T exists in the direction parallel to the axis of the loop. The coil is now rotated about a diameter through an angle of 60°. The operation takes 0.100 s (a) Find the average emf induced in the coil. (b) If the coil is a closed one (with the two ends joined together) and has a resistance of 4.00 Ω, calculate the net charge crossing a cross-section of the wire of the coil.

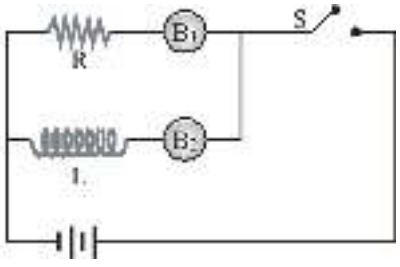
(1) (a) $2\pi \times 10^{-2}$V, (b) $\pi/2 \times 10^{-3}$ C

(2) (a) $2\pi \times 10^{-2}$V, (b) $\pi \times 10^{-3}$ C

(3) (a) $3\pi \times 10^{-2}$V, (b) $\pi \times 10^{-2}$ C

(4) (a) $4\pi \times 10^{-2}$V, (b) $2\pi \times 10^{-2}$ C

10) The adjoining figure shows two bulbs B_1 and B_2 resistor R and an inductor L. Initially switch S is closed. When the switch S is turned off



(1) Both B_1 and B_2 die out promptly

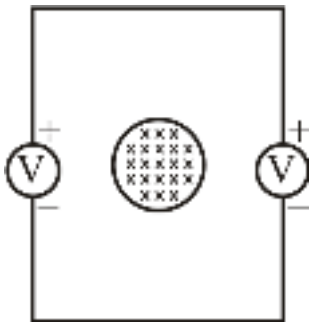
(2) Both B_1 and B_2 die out with some delay

(3) B_1 dies out promptly but B_2 with some delay

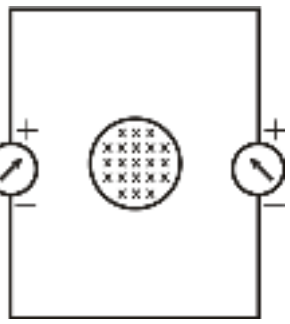
(4) B_2 dies out promptly but B_1 with some delay

11) Two similar voltmeters with large resistance are connected in a loop as shown. Inside the loop, there is a cylindrical region through which magnetic flux is distributed uniformly. The magnetic field increases uniformly with time. The voltmeters are bidirectional i.e. the needle deflects to right if current in it flows from + terminal to – terminal and needle deflects to left if current in it flows from

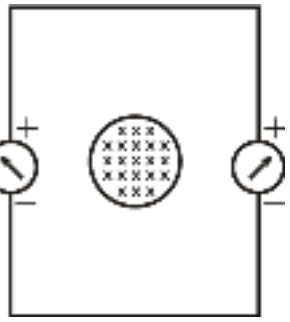
– terminal to + terminal. Which of the following figures shows the readings of voltmeters correctly?



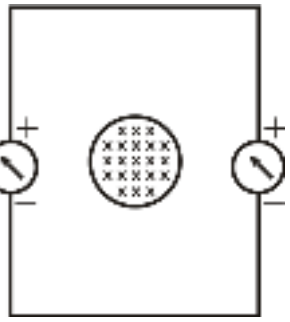
(1)



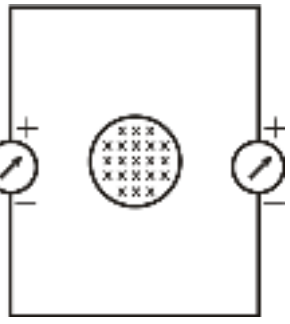
(2)



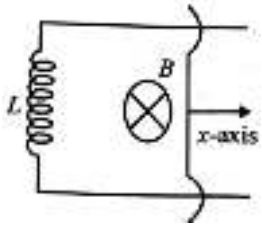
(3)



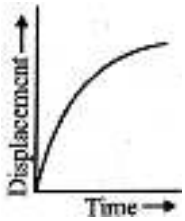
(4)

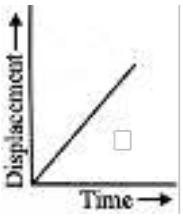


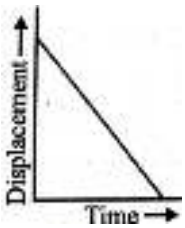
12) A coil of self inductance L is connected at one end of two rails as shown in figure. A connector of length □, mass m can slide freely over the two parallel rails. The entire set up is placed in a magnetic field of induction B going into the page. At an instant t = 0 an initial velocity v_0 is imparted to it and as a result of that it starts moving along x-axis. The displacement of the connector is represented by

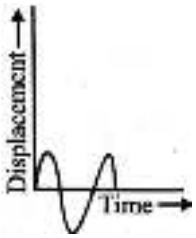


the figure.

(1) 

(2) 

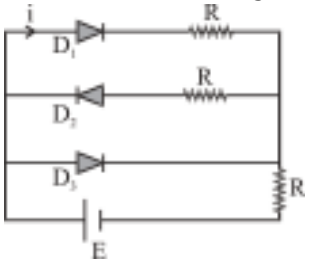
(3) 

(4) 

13) GaAs(ΔEg = 1.5 eV) as an LED can emit

(1) Blue light
(2) Green light
(3) UV rays
(4) Infrared rays

14) In the following circuit of PN junction diodes D_1, D_2 and D_3 are ideal then i is :-



(1) E/R
(2) E/2R
(3) 2E/3R

(4) Zero

15) In given boolean expression Y equal to
Y = A.B + (A + B) . A :-

(1) A

(2) B

(3) A.B

(4) A + B

16) Four independent waves are represented by the equations
$y_1 = a_1 \sin \omega t$, $y_2 = a_2 \sin \omega t$,

$y_3 = a_3 \cos \omega t$, $y_4 = a_4 \sin(\omega t + \frac{\pi}{3})$

Then the waves for which phenomenon of interference will be observed are :–

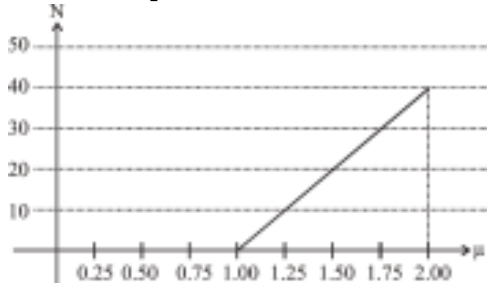
(1) 1 and 3

(2) 1 and 4

(3) all 1, 2, 3 and 4

(4) none

17) The slits in a double-slit interference experiment are illuminated by orange lights (λ = 600 nm). A thin transparent plastic of thickness t is placed in front of one of the slits. The number of fringes (N) shifting on screen is plotted versus the refrective index (μ) of the plastic in graph shown. The value of t is :-



(1) 4.8 mm

(2) 48 μm

(3) 2.4 μ m

(4) 24 μm

18) If the source of light used in a Young's double slit experiment is changed from red to violet :

(1) consecutive fringe lines will come closer.

(2) the central bright fringe will become a dark fringe.

(3) the fringes will become brighter .

(4) the intensity of minima will increase.

19) Unpolarized light of intensity I passes through an ideal polarizer A. Another identical polarizer B is placed behind A. The intensity of light beyond B is found to be $\frac{I}{2}$. Now another identical polarizer

C is placed between A and B. The intensity beyond B is now found to be $\frac{I}{8}$. The angle between polarizer A and C is :

(1) 30°

(2) 45°

(3) 60°

(4) 0°

20) **Assertion (A) :** Light radiation exerts pressure on a surface.
Reason (R) : Light radiation possess linear momentum.

(1) Both (A) and (R) are correct and (R) is correct explanation of (A)

(2) Both (A) and (R) are correct and (R) is not correct explanation of (A)

(3) Both (A) & (R) incorrect

(4) (A) incorrect while (R) correct

21) Energy flux of 50 mWm^{-2} incident normally on a perfectly reflective surface of area 9 mm^2 find magnitude of average force experienced by the surface :-

(1) 1.5×10^{-17} N

(2) 4.5×10^{-15} N

(3) 3×10^{-15} N

(4) 5.2×10^{-10} N

22) The work functions of metals A and B are in the ratio 1 : 2, If light of frequencies f and 2f are incident on the surface of A and B respectively, the ratio of the maximum kinetic energies of photoelectrons emitted is [f is greater than threshold frequency of A, 2f is greater than threshold frequency of B]

(1) 1 : 1

(2) 1 : 2

(3) 1 : 3

(4) 1 : 4

23) If energy of an electron is increased by 69% find percentage change in its de-Broglie wave length (appoximate) :-

(1) 25% decrease

(2) 23% decrease

(3) 64% decrease

(4) 16% decrease

24) A proton is accelerated through a potential difference of 100 volt. To have same de-Broglie wavelength what potential difference must be applied across doubly ionised $^{7}_{3}Li$ atom :-

(1) 8.52 volt

(2) 3.16 volt

(3) 7.14 volt

(4) 3.38 volt

25) A nucleus ${}^{A}_{Z}X$ decays by emitting two alpha particles, $2\beta^-$ and $4\beta^+$ then ratio of number of neutrons to number of protons in the final nucleus :

(1) $\frac{A-Z-9}{Z-3}$

(2) $\frac{A-Z-1}{Z-8}$

(3) $\frac{A-Z-2}{Z-6}$

(4) $\frac{A-Z+1}{Z-6}$

26) A nucleus of ${}^{210}_{84}Po$ originally at rest emits α-particle with speed V . What will be the recoil speed of the daughter nucleus :-

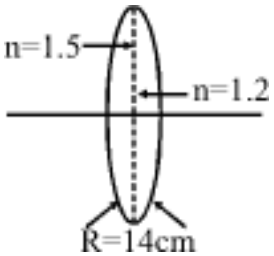
(1) $\frac{4V}{206}$

(2) $\frac{4V}{214}$

(3) $\frac{V}{206}$

(4) $\frac{V}{214}$

27) A bi-convex lens is formed with two thin plano-convex lenses as shown in the figure. Refractive index n of the first lens is 1.5 and that of the second lens is 1.2. Both the curved surface are of the same radius of curvature R = 14 cm. For this bi-convex lens, for an object distance of 40 cm, the image distance will be :-



(1) –280.0 cm

(2) 40.0 cm

(3) 21.5 cm

(4) 13.5 cm

28) A thin linear object of size 1mm is kept along the principal axis of a convex lens of focal length 10cm. The object is at 15 cm from the lens. The length of image is

(1) 1 mm

(2) 2 mm

(3) 3 mm

(4) 4 mm

29) In a simple microscope we use lens of focal length 5 cm and final image is formed at 25 cm away from the lens. Find out magnifying power of simple microscope.

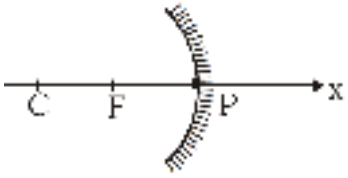
(1) 5

(2) 5.5

(3) 6

(4) 11

30) A concave mirror is kept as shown in figure. Its principal axis coincides with the x–axis. A real object is moving towards the pole of the mirror starting from the –∞. In which region image will never form :–



(1) Between –∞ and C

(2) Between C and F

(3) Between F and P

(4) Between P and + ∞

31) A light ray is travelling from air to a liquid of refractive index $\sqrt{3}$. If the incident beam is rotated at a rate of 3 rad/sec then the angular speed of refracted beam when angle of incidence is 60° is :-

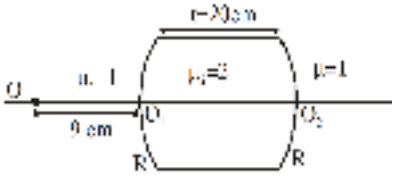
(1) 3 rad/sec

(2) $\sqrt{3}$ rad/sec

(3) 1 rad/sec

(4) $3\sqrt{3}$ rad/sec

32) In the given figure, if object is placed at a distance 9 cm from O_1 then find the image distance (in cm) from O_2. (R = 9 cm for both surface)



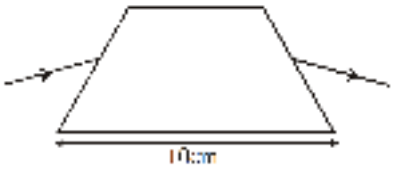
(1) 9 cm

(2) 20 cm

(3) 18 cm

(4) Infinity

33) Consider an isosceles trapezoid of material of refractive index $\sqrt{2}$ and dimensions of sides being

5cm, 5cm, 10 cm and 5cm. The angle of minimum deviation by this arrangement when light is incident from air and emerges in air is 

(1) 22.5°

(2) 30°

(3) 45°

(4) 60°

34) A beam of light is converging towards a point on the screen. If a plane parallel plate of glass of refractive index μ and thickness t is introduced in the path of the beam, the convergent point is shifted

(1) $t\left[1 - \frac{1}{\mu}\right]$ away

(2) $t\left[1 + \frac{1}{\mu}\right]$ nearer

(3) $t\left[1 + \frac{1}{\mu}\right]$ away

(4) $t\left[1 - \frac{1}{\mu}\right]$ nearer

35) A diver in a lake wants to signal his distress to a person sitting on the edge of the lake flashing his water proof torch. He should direct the beam

(1) vertically upwards

(2) horizontally

(3) at an angle to the vertical which is slightly less than the critical angle

(4) at an angle to the vertical which is slightly more than the critical angle

36) A convex lens, made of a material of refractive index 1.5 and having a focal length of 10 cm, is immersed in a liquid of refractive index 3.0. The lens will behave as a

(1) converging lens of focal length 10 cm

(2) diverging lens of focal length 10 cm

(3) converging lens of focal length 10/3 cm

(4) diverging lens of focal length 30 cm

37) Arvind need to identify the electrical components from group of components labelled as A, B, C and D. He performed some simple tests and obtained following observations :
(a) A has silver ring at its one end, with two terminals. It can conduct electricity only in one direction.
(b) B is connected to d.c. source, but current gradually decreases and becomes zero. It is two terminal device and behaves same in both directions.
(c) C is a two terminal device, which conduct electricity only in one direction, with emitting light.

(d) D is two terminal device with several coloured rings on it. It conducts current in both directions with same steady current.
The correct identification of A, B, C and D are respectively :

(1) Resistor, Photodiode, Diode, Capacitor

(2) Diode, LED, Capacitor, Resistor

(3) Diode, Capacitor, LED, Resistor

(4) LED, Capacitor, Diode, Resistor

38) Which of the following statements about P – N junction diode is incorrect:

(1) The forward bias resistance of a P–N junction diode is very high before knee voltage and becomes very low after certain voltage greater than knee voltage.

(2) The reverse current in an ideal diode can be minimum or constant before breakdown.

(3) The reverse current becomes infinity at breakdown.

(4) In avalanche breakdown, pairs of electron and holes are produced due to collision of electrons with covalent bonds and its thermal effect.

39) A travelling microscope whose main scale has 20 divisions in 1 cm. is used to measure refractive index of a medium. Also, its 50 vernier scale divisions are equal to 49 main scale divisions. The readings obtained while measuring refractive index of glass slab are :

	Position of T. M.	**MSR**	**VSR (Coinciding div)**
(a)	When microscope is focussed on mark without slab	3.20 cm	4th
(b)	When microscope is focussed on mark with slab.	4.60 cm	47th
(c)	When microscope is focussed on slab surface sprinkled with lycopodium powder	8.25 cm	4th

The refractive index of glass slab is :-

(1) 1.2

(2) 1.4

(3) 1.6

(4) 1.65

40) For finding out focal length of a convex lens using two needles by parallax method, object and image needles are 25 cm apart. If focal length of lens is found 6 cm then possible values of object needle distance from lens are :

(1) 10 cm and 15 cm

(2) 12 cm and 13 cm

(3) 8 cm and 17 cm

(4) 5 cm and 20 cm

41) In a circuit for finding the resistance of a galvanometer by half deflection method, a 6 volt battery and a high resistance of 11 kΩ are used. The figure of merit of the galvanometer is 60 μA/division. In the absence of shunt resistance, the galvanometer produces a deflection of θ = 9 divisions when current flows in the circuit. The value of shunt resistance that can cause deflection of θ/2 is closest to :

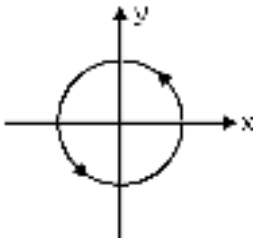
(1) 55 Ω

(2) 110 Ω

(3) 220 Ω

(4) 550 Ω

42) Current i = 2.5 A flows along the circle $x^2 + y^2 = 9\ cm^2$ (here x & y in cm) as shown. Magnetic field at point (0, 0, 4 cm) is



(1) $\left(36\pi \times 10^{-7}T\right)\hat{k}$

(2) $\left(36\pi \times 10^{-7}T\right)\left(-\hat{k}\right)$

(3) $\left(\frac{9\pi}{5} \times 10^{-7}T\right)\hat{k}$

(4) $\left(\frac{9\pi}{5} \times 10^{-7}T\right)\left(-\hat{k}\right)$

43) A particle having some charge is projected in x-y plane with a speed of 5 m/s in a region having uniform magnetic field along z-axis. Which of the following cannot be the possible value of velocity at any time ?

(1) $3\hat{i} - 4\hat{j}$

(2) $4\hat{i} + 3\hat{j}$

(3) $5\hat{j}$

(4) $3\hat{i} - 4\hat{k}$

44) The Q factor of a series LCR circuit with L = 2H, C = 32μF and R = 10 Ω is :-

(1) 15

(2) 20

(3) 25

(4) 30

45) The electric field of an electromagnetic wave is given by $E = 50\sin\omega\left(t - \frac{x}{c}\right)$ N/C. The energy

contained in a cylinder of cross-section of 10 cm^2 and length 50 cm along X-axis will be _____

(1) 5.5×10^{-12} J
(2) 5.5×10^{-6} J
(3) 1.1×10^{-11} J
(4) 1.1×10^{-5} J

CHEMISTRY

1) Match List-I with List-II

List-I		List-II	
(a)	$[Fe(NH_3)_6]^{+2}$	(P)	3.9 BM
(b)	$[Co(H_2O)_6]^{+3}$	(Q)	2.8 BM
(c)	$[Ni(H_2O)_6]^{+2}$	(R)	0 BM
(d)	$[Cr(CN)_6]^{-3}$	(S)	4.9 BM

Select correct options & answer(s)
(1) a-R, b-S, c-Q, d-P
(2) a-S, b-R, c-P, d-Q
(3) a-S, b-R, c-Q, d-P
(4) a-R, b-S, c-P, d-Q

2) Consider the following statements
S1 → B_2F_6 doesn't exist
S2 → $N(SiH_3)_3$ work as a good Lewis base
S3 → Boric acid does not show back bonding
Option & Answer -

(1) S1, S2 and S3 are correct
(2) S1 and S2 are correct
(3) S1, S2 and S3 are incorrect
(4) S2 and S3 are incorrect

3) Consider the following statements
S1 → Sn^{+2} is more stable than Pb^{+2}
S2 → Bi does not form any compound with +5 charge due to inert pair effect
S3 → Na_2HPO_2 is exist as a acidic salt.
Options and answer -

(1) S1, S2 and S3 correct
(2) S2, S3 correct
(3) S1, S3 incorrect
(4) S1, S2, S3 incorrect

4) **Statement-I :** $[MnCl_6]^{-3}$, $[FeF_6]^{-3}$ and $[CoF_6]^{-3}$ are outer orbital complexes involving sp^3d^2 hybridisation and are paramagnetic corresponding four, five and four unpaired e^-.
Statement-II : $[Co(C_2O_4)_3]^{-3}$ is outer orbital complex involving sp^3d^2 hybridisation and four unpaired e^-.

(1) **Statement I** is correct & **Statement II** is incorrect.

(2) **Statement I** is incorrect & **Statement II** is correct.

(3) Both **Statement I** and **Statement II** are incorrect.

(4) Both **Statement I** and **Statement II** are correct.

5) Identified incorrect statement -

(1) If $\Delta_0 < P$, the fourth electron enters one of the eg orbital giving the configuration $t_{2g}^3\ eg^1$

(2) If $\Delta_0 > P$, forth electron enters one of the t_{2g} orbital giving the configuration $t_{2g}^4\ eg^0$

(3) Calculation show d^4 to d^7 coordination entities are less stable for strong field as compared to weak field.

(4) In tetrahedral coordination entity formation the d orbital splitting is inverted and is smaller as compared to octahedral field splitting.

6) Ni^{+2} + 2DMG → ''A'' complex
Select the correct statement regarding complex A.

(1) A complex is colourless

(2) A doesn't show hydrogen bonding

(3) A show ion-dipole force

(4) A is unstable complex

7) Match the column

Identification Test		**Colour**	
(i)	Cu^{+2} with NH_3	(a)	Violet
(ii)	Na_2S with sodium nitroprusside	(b)	Crimson red
(iii)	$FeCl_2$ + NO (Solution)	(c)	Blue
(iv)	Flame test of Lithium	(d)	Brown

Options :

(1) (i) - a, (ii) - b, (iii) - d, (iv) - c

(2) (i) - b, (ii) - c, (iii) - a, (iv) - d

(3) (i) - d, (ii) - c, (iii) - a, (iv) - b

(4) (i) - c, (ii) - a, (iii) - d, (iv) - b

8) Identified correct statements -

(1) The electronic configurations of outer orbitals of Zn, Cd, Hg and Rg are represented by the general formula (n – 1) d^{10} ns^2.

(2) In 3d series scandium (II) is virtually unknown

(3) In group 6 Mo (VI) and W (VI) are found to be less stable than Cr(VI)

(4) Cr^{+2} is oxidising agent and Mn^{+3} is reducing agent when both have d^4 configuration

9)

List-I (Basic radicals)		**List-II (Group)**	
(P)	Pb^{+2}	(1)	Group-III
(Q)	Fe^{+3}	(2)	Group-I
(R)	Ni^{+2}	(3)	Group-VI
(S)	Mg^{+2}	(4)	Group-IV

Correct matching is

	P	Q	R	S
(1)	2	1	3	4
(2)	1	3	2	4
(3)	2	1	4	3
(4)	1	2	3	4

(1) 1

(2) 2

(3) 3

(4) 4

10) Which is incorrect statement

(1) H_2S can decolourise acidified $KMnO_4$

(2) $K_2Cr_2O_7$ can react with H_2O_2 and give pink colour CrO_5 compound

(3) $KMnO_4$ is less soluble in water

(4) $K_2Cr_2O_7$ (orange) becomes K_2CrO_4 (yellow) in basic medium

11) Before adding the reagent of group III, if some drop of conc. HNO_3 doesn't added then?

(1) $Fe(OH)_3$ precipitate will form easily

(2) $Al(OH)_3$ precipitate will form

(3) Fe^{+2} will not oxidise into Fe^{+3}

(4) HNO_3 must be required to supporting role for dil HCl

12) Select the correct match in flame test -

	Metals	Colour in flame
(1)	Na	Brick red
(2)	K	Violet
(3)	Ca	Golden yellow
(4)	Ba	Crimson red

(1) 1

(2) 2

(3) 3

(4) 4

13) Choose the incorrect statement

(1) In Pnicogen family bismuth hardly forms any compound in – 3 oxidation state

(2) In case of nitrogen, all oxidation states from +1 to +4 tend to disproportionate in acidic solution

(3) Nitrogen has unique ability to form pπ-dπ multiple bonds with itself with other elements having small size and high EN [i.e. C, O]

(4) The single N–N bond is weaker than the single P-P bond because of high interelectronic repulsion of the non-bonding electrons.

14) Choose incorrect match

	Group	Oxide formation
(1)	Boron's family	E_2O_3
(2)	Nitrogen's family	E_2O_5
(3)	Carbon's family	E_2O_7
(4)	Halogen's family	E_2O

here E means central atom like B,C,N,X.

(1) 1

(2) 2

(3) 3

(4) 4

15) Find out correct statement

(1) Bromine oxides like Br_2O, BrO_2, BrO_3 are highly stable due to middle row anomaly

(2) Use of I_2O_5 in the estimation of carbon monoxide

(3) Iodine give yellow colour with starch

(4) Composition of bleaching powder is $CaCl_2.Ca(OH)_2.2H_2O$

16) **Assertion (A) :** Cr has $3d^5 4s^1$ electronic configuration instead of $3d^4\ 4s^2$
Reason (R) : The energy gap between the two sets (3d and 4s) of orbitals is small enough to prevent electron entering the 3d orbitals.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is true but **Reason** is false.

(4) **Assertion** is false but **Reason** is true.

17) Which of the following pair of transition elements exhibit highest and lowest density.

(1) Os and Ir

(2) Os and Pt

(3) Hg and Sc

(4) Ir and Sc

18)

The "f" orbitals are half and completely filled respectively in Lanthanoid ions.
[Given → Atomic no. Eu = 63, Sm = 62, Tm = 69, Tb = 65, Yb = 70, Dy = 66]

(1) Eu^{+2} and Tm^{+2}

(2) Sm^{+2} and Tm^{+2}

(3) Tb^{+4} and Yb^{+2}

(4) Dy^{+2} and Yb^{+2}

19) When Cu^{+2} ion is treated with KI a white precipitate X appears in solution, the solution is titrated with sod. thiosulphate the compound Y is formed, X and Y respectively are :

(1) X = Cu_2I_2, Y = $Na_2S_4O_5$

(2) X = Cu_2I_2, Y = $Na_2S_4O_6$

(3) X = CuI_2, Y = Na_2S_4O3

(4) X = CuI_2, Y = Na_2S_4O4

20) When 0.1 mole $CoCl_3$. $5NH_3$ is treated with an excess of $AgNO_3$, 0.2 mole of AgCl are obtained, the conductivity of the solution will corresponds to -

(1) 1 : 3 electrolyte

(2) 1 : 2 electrolyte

(3) 1 : 1 electrolyte

(4) 3 : 1 electrolyte

21) When I_2 is passed through KCl, KF and KBr

(1) Cl_2 and Br_2 are evolved

(2) Only Cl_2 evolved

(3) Cl_2, F_2 and Br_2 evolved

(4) None of these

22) Which of the following doesn't have a metal carbon bond -

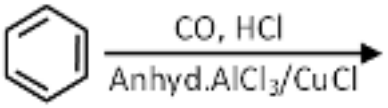
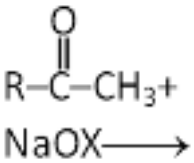
(1) $[Ni(CO)_4]$

(2) $[Al(OC_2H_5)_3]$

(3) C_2H_5MgBr

(4) $K[Pt(C_2H_4)Cl_3]$

23) Match **List-I** with **List-II**.

List-I		**List-II**	
(a)		(i)	Williamson synthesis
(b)		(ii)	Gattermann-Koch Reaction
(c)	$R–CH_2–OH + R'COOH \xrightarrow{Conc.\ H_2SO_4}$	(iii)	Haloform reaction
(d)	$RO^{\ominus}\ Na^{\oplus} + R – X$	(iv)	Esterification

Choose the **correct** answer from the options given below.

(1) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

(2) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

(3) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)

(4) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

24) The correct order of following compounds towards S_N2 mechanism is :-

(1) $C_6H_5C(CH_3)(C_6H_5)Br > C_6H_5CH(C_6H_5)Br > C_6H_5CH(CH_3)Br > C_6H_5CH_2Br$

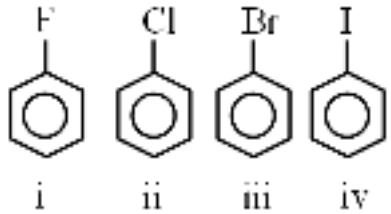
(2) $C_6H_5C(CH_3)(C_6H_5)Br > C_6H_5CH_2Br > C_6H_5CH(CH_3)Br > C_6H_5CH(C_6H_5)Br$

(3) $C_6H_5CH_2Br > C_6H_5C(CH_3)(C_6H_5)Br > C_6H_5CH(CH_3)Br > C_6H_5CH(C_6H_5)Br$

(4) $C_6H_5CH_2Br > C_6H_5–CH(CH_3)Br > C_6H_5CH(C_6H_5)Br > C_6H_5C(CH_3)(C_6H_5)Br$

25)

Give order of rate of ESR (Electrophilic substitution reaction)



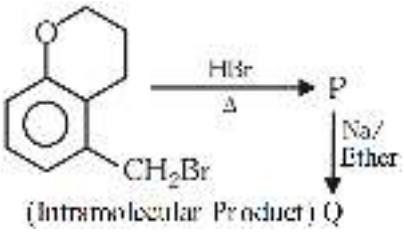
(1) (i) > (ii) > (iii) > (iv)

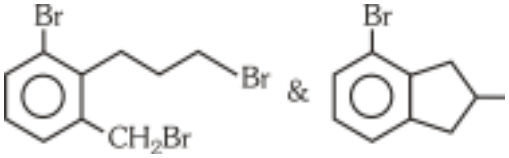
(2) (iv) > (iii) > (ii) > (i)

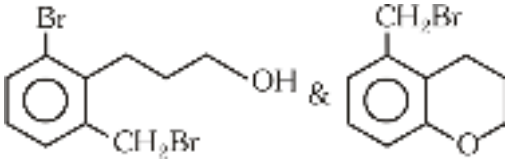
(3) (ii) > (i) > (iii) > (iv)

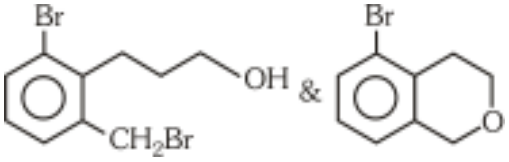
(4) (ii) > (i) > (iv) > (iii)

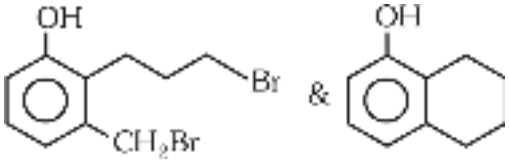
26) In the following reaction sequence, structures of P and Q, are respectively



(1) 

(2) 

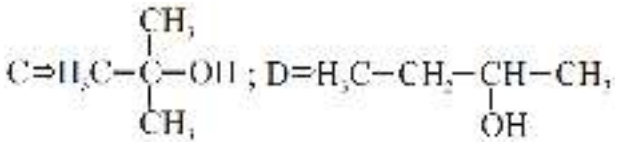
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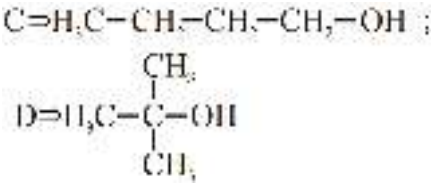
(4) 

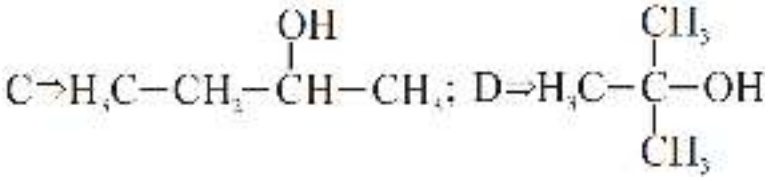
27) Two compounds A and B with same molecular formula C_3H_6O undergo Grignard's reaction with methylmagnesium bromide followed by acidic hydrolysis to give products C and D. Products C and D show following chemical tests.

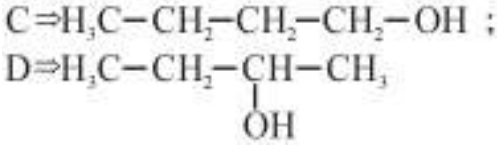
Test	C	D
Ceric ammonium nitrate Test	Positive	Positive
Lucas Test	Turbidity obtained after five minutes	Turbidity obtained immediately
Iodoform Test	Positive	Negative

C and D respectively are :

(1) 

(2) 

(3) 

(4) 

28) Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : Chlorine is an electron withdrawing group but it is ortho, para directing in electrophilic aromatic substitution.

Reason (R) : Inductive effect of chlorine destabilises the intermediate carbocation formed during the electrophilic substitution, however due to the more pronounced resonance effect, the halogen stabilises the carbocation at ortho and para positions. In the light of the above statements, choose the most appropriate answer from the options given below :

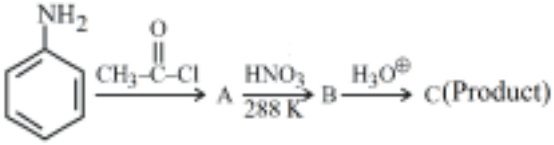
(1) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

(2) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

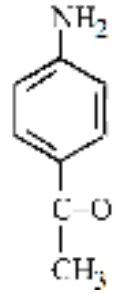
(3) **(A)** is correct but **(R)** is not correct

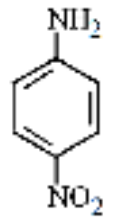
(4) **(A)** is not correct but **(R)** is correct.

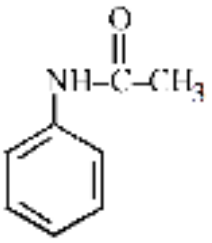
29) Suppose following reaction

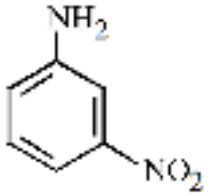


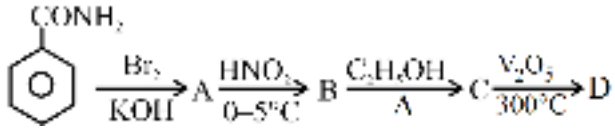
The major product 'C' will be :-

(1) 

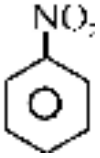
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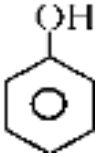
(3) 

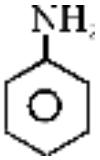
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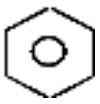
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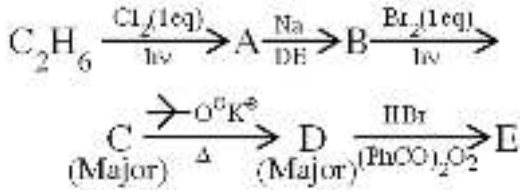
What is D ?

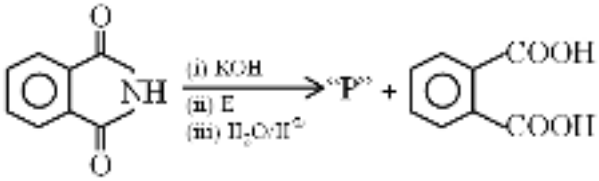
(1) 

(2) 

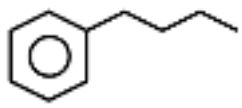
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(4) 

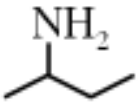
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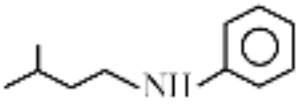


Final major product "P" in above reaction sequence is :

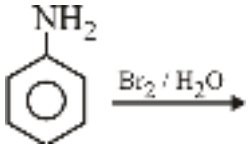
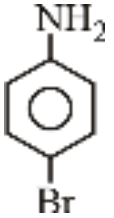
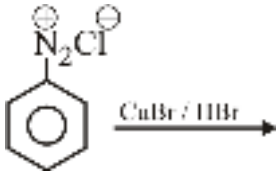
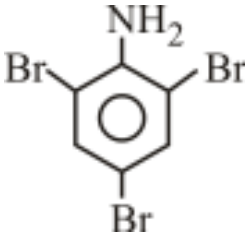
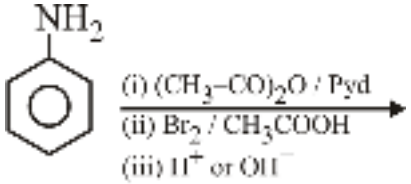
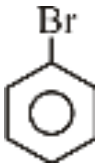
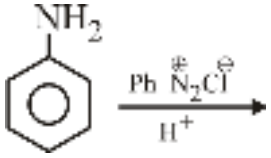
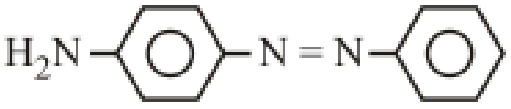
(1) 

(2) 

(3) 

(4) 

32) Match the column :-

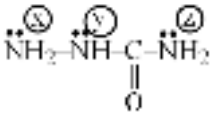
	Column-I (Reaction)		Column-II (Product)
(A)		(p)	
(B)		(q)	
(C)		(r)	
(D)		(s)	

(1) A → q, B → r, C → p, D → s

(2) A → r, B → q, C → p, D → s

(3) A → s, B → p, C → q, D → s

(4) A → p, B → q, C → r, D → s

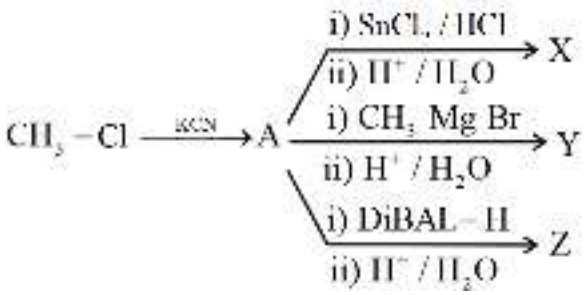
33) Which N of semicarbazide donate □.p & behave like N$\overset{\ominus}{u}$. 

(1) x

(2) y

(3) z

(4) y & z



34)
Identify incorrect statement

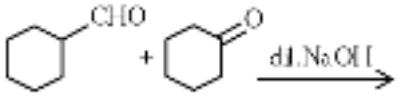
(1) X, Y, Z all give iodoform test

(2) X, Y, Z all give Tollen's test

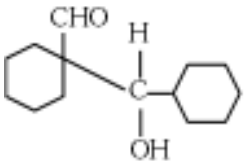
(3) A is methyl cyanide

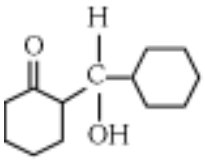
(4) Rate of Nucleophilic Addition X > Y

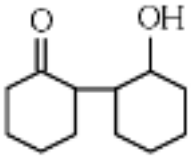
35) Find major product

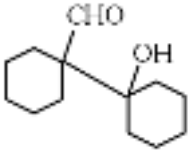


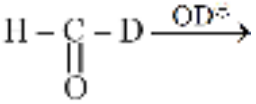
major product

(1) 

(2) 

(3) 

(4) 

36) 

Major product.

(1) $D–COO^{\ominus}$ & $CH_2D–OH$

(2) $D–COO^{\ominus}$ & $CH_2D–O^{\ominus}$

(3) $H–COO^{\ominus}$ & $CD_2H–OD$

(4) $D\text{–}COO^{\ominus}$ & $CH_2D\text{–}OD$

37) **Assertion :** The boiling points of aldehydes and ketones are higher than hydrocarbons and ethers of comparable molecular masses.
Reason : In aldehydes and ketones arising out of the dipole-dipole interactions.

(1) Assertion is correct, reason is correct, reason is a correct explanation for assertion.

(2) Assertion is correct, reason is correct, reason is not a correct explanation for assertion

(3) Assertion is correct, reason is incorrect

(4) Assertion is incorrect, reason is correct

38) Which of the following compound gives white crystals with $NaHSO_3$:-

(1) Acetophenone

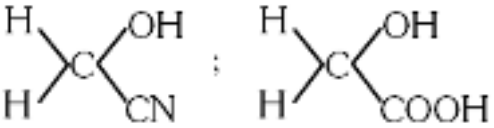
(2) Benzophenone

(3) Butanone

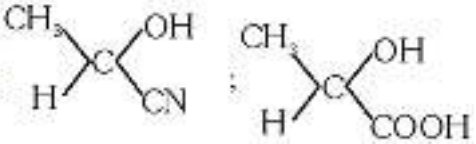
(4) 3-Pentanone

39) A carbonyl compound 'A' reacts with hydrogen cyanide to form a cyanohydrin 'B' which on hydrolysis gives an optically active alpha hydroxy acid 'C'. 'A' gives a positive iodoform test, 'A', 'B' and 'C' are given by the set :-

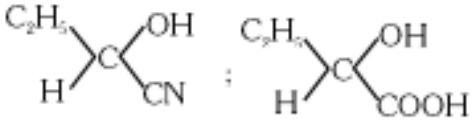
(1) HCHO;



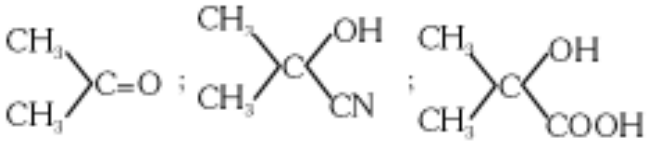
(2) CH_3CHO;



(3) CH_3CH_2CHO;



(4)



40) Match column-I with column-II

Column-I		Column-II	
(I)	Dumas method	(a)	$AgNO_3$
(II)	Kjeldahl's method	(b)	Silica gel
(III)	Carius method	(c)	Nitrogen gas
(IV)	Chromatography	(d)	Ammonium Sulphate

(1) I→c, II→d, III→a, IV→b

(2) I→c, II→a, III→d, IV→b

(3) I→c, II→b, III→a, IV→d

(4) I→d, II→c, III→a, IV→b

41) **Statement-I :** Thin layer chromatography is an adsorption chromatography.
Statement-II : A thin layer of silica gel is spread over a glass plate of suitable size in thin layer chromatography which acts as an adsorbent.

(1) Statement-I & statement-II are incorrect

(2) Statement-I is true & II is false

(3) Statement-I is false & II is true

(4) Statement-I & II both are correct

42) During estimation of nitrogen present in an organic compound, the ammonia evolved from 0.5 g of the compound in kjeldahl's estimation of nitrogen, neutralized 10 mL of 1 MH_2SO_4. The percentage of nitrogen in the compound is :

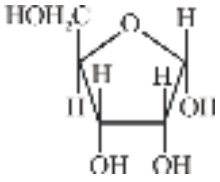
(1) 46%

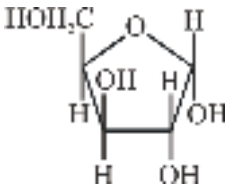
(2) 51.0%

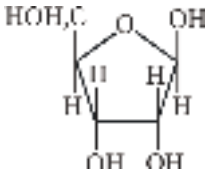
(3) 56%

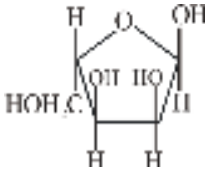
(4) 49%

43) Which of the following compound is anomer of given compound ?



(1) 

(2) 

(3) 

(4) All of the above

44) In which of the following properties of an open chain structure of glucose could not be explained by Baeyer –

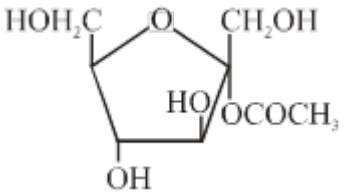
(1) Glucose contain aldehyde group, but does not give schiff's test

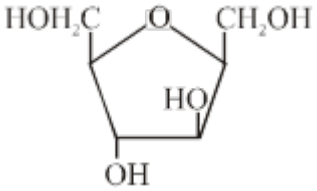
(2) Glucose does not react with sodium hydrogen sulphite and ammonia

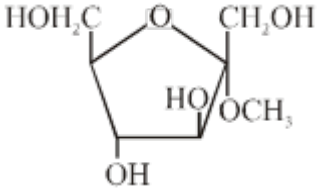
(3) The pentaacetate of glucose does not react with hydroxyl amine indicates absence of –CHO group

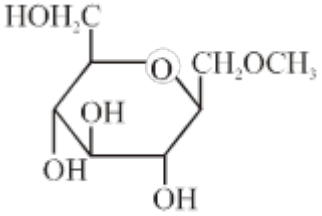
(4) All of these

45) Which of the following compounds will behave as a reducing sugar in an aqueous KOH solution ?

(1) 

(2) 

(3) 

(4) 

BIOLOGY

1) Select the incorrect statement w.r.t. incomplete dominance.

(1) Phenotype of F_1 do not resemble either of the two parents

(2) Genotypic and phenotypic ratio are dissimilar

(3) The Genotypic ratio is 1 : 2 : 1

(4) The inheritance of flower colour in the dog flower is a good example.

2) Rediscovery of Mendal's work done by :

(1) Sutton and Boveri

(2) Morgan and de vries

(3) de Vries, Correns and Tschermak

(4) Correns and Morgan

3) Pedigree is not utilised to trace the inheritance of :-

(1) Specific trait

(2) Abnormality

(3) Disease

(4) Sex ratio

4) Test cross is used for the determination of genotype, of :-

(1) Homozygous dominant plant

(2) Heterozygous dominant plant

(3) Homozygous recessive plant

(4) (1) and (2) both

5) Which of the following statement is not correct ?

(1) The pattern of inheritance of chromosomal disorder can be traced in a family by the pedigree analysis.

(2) Cystic fibrosis is an autosomal recessive disorder.

(3) Colourblindness occurs in about 8 percent of males.

(4) The possibility of a female becoming a haemophilic is extremely rare.

6) Match column I with column II and select the correct option from the codes given below -

	Column-I		**Column-II**
A	A single trait controlled by three or more than three allele	i	Pleiotropy
B	A single trait controlled by three or more than three genes	ii	Multiple alleles
C	A single gene exhibits multiple phenotypic expression	iii	Polygenic inheritance

(1) A - (ii), B - (iii), C - (i)

(2) A - (iii), B - (ii), C - (i)

(3) A - (i), B - (ii), C - (iii)

(4) A - (ii), B - (i), C - (iii)

7) If the genotypes of husband and wife are I^AI^B and I^AI^O : Among the blood types of their children, how many different genotypes and phenotypes are possible?

(1) 2 genotypes, 3 phenotypes

(2) 3 genotypes, 4 phenotypes

(3) 4 genotypes, 4 phenotypes

(4) 4 genotypes, 3 phenotypes

8) Given below are two Statements :-
Statement-I :- Henking (1891) could trace a specific nuclear structure all through spermatogenesis in a few humans.
Statement-II :- Both in human and in *Drosophila*, the female has a pair of XX chromosomes (homogametic) and the male XY (heterogametic) composition.

Choose the most appropriate answer from given below options :-

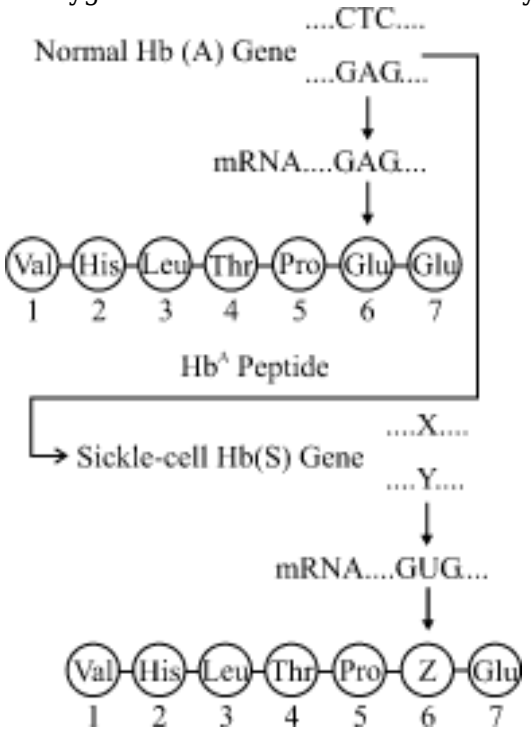
(1) Both Statement I and Statement II are correct

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are incorrect

9) Sickle-cell anaemia is an autosomal linked recessive trait that can be transmitted from parents to the offspring when both the partners are carrier or heterozygous. The disease is controlled by a single pair of allele, Hb^A and Hb^S. Identify X, Y and Z:-



(1) GTG, GAC, Val

(2) CAC, GTG, Val

(3) GTA, GAG, Val

(4) GTC, GAC, Val

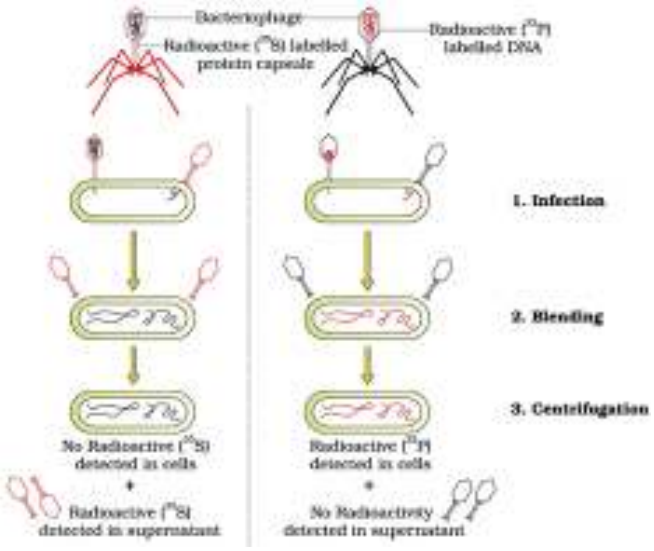
10) Pedigree analysis is applicable for how many of the following disorders :
Turner syndrome, Down syndrome,
Cystic fibrosis, Albinism, Haemophilia,
Colorblindness, Phenylketonuria

(1) 3

(2) 4

(3) 5

(4) 6

11) 

Given below are two statements about the above experiment :
Statement-I :- The unequivocal proof that DNA is the genetic material come from this experiment which performed by Avery, Macleod and McCarty.
Statement-II :- This experiment indicates that proteins did not enter the bacteria from the viruses.
In the light of the above statements
Choose the most appropriate answer from the option given below :

(1) Statement-I is correct but statement-II is incorrect

(2) Statement-I is incorrect but statement-II is correct

(3) Both statement-I & statement-II are incorrect

(4) Both statement-I & statement-II are correct

12) In the study of the nature of the genetic code in a bacterium, a researcher observed that the bacterial m-RNA contains only 3 types of nitrogenous bases. How many minimum codons are possible that can code for all 20 types of amino acids in this bacterium :

(1) 64

(2) 9

(3) 3

(4) 27

13) **Assertion (A) :** In eukaryotes, the primary transcript (hnRNA) must undergo splicing, capping, and polyadenylation before becoming functional mRNA, because raw transcripts cannot directly participate in translation.
Reason (R) : These post-transcriptional modifications protect RNA from exonuclease degradation, enhance transport from nucleus to cytoplasm, and promote efficient ribosome recognition during protein synthesis.

(1) Both A and R are true, and R is the correct explanation of A.

(2) Both A and R are true, but R is not the correct explanation of A.

(3) A is true, but R is false.

(4) A is false, but R is true.

14) Select the incorrect statement :-

(1) The promoter and terminator flank the structural gene in a translation unit.

(2) A gene is defined as the functional unit of inheritance.

(3) Expression of a character is affected by promoter and regulatory sequence of structural gene.

(4) All the reference point, while defining a transcription unit, is made with coding strand.

15) Match List-I with List-II :

	List-I		**List-II**
(a)	Histone	(i)	Loosely packed chromatin
(b)	Nucleosomes	(ii)	DNA wrapped around histone octamer
(c)	Euchromatin	(iii)	Tightly packed chromatin
(d)	Heterochromatin	(iv)	Positive charge basic protein

(1) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)

(2) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)

(3) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)

(4) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

16) **Assertion (A) :** In eukaryotes, transcription and translation cannot be coupled.
Reason (R) : Eukaryotic mRNA requires extensive processing, and transcription occurs in the nucleus while translation occurs in the cytoplasm.

(1) Both A and R are true but R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

17) **Statement-1 :** By switching the position of promoter with terminator, the definition of coding and template strands could be reversed.
Statement-2 : The structural gene in a transcription unit could be said as monocistronic (mostly in eukaryotes) or polycistronic (mostly in prokaryotes).

(1) Both statement-1 and statement-2 are correct.

(2) Both statement-1 and statement-2 are incorrect.

(3) Statement-1 is correct but statement-2 is incorrect.

(4) Statement-1 is incorrect but statement-2 is correct.

18) Calculate the number of nucleosomes in human sperm ?

(1) 1.65×10^9

(2) 3.3×10^7

(3) 1.65×10^7

(4) 3.3×10^9

19) Which of the following is the goal of H.G.P. ?

(1) Identify all the gene in human DNA.
(2) Determine the sequences of the 3 billion chemical base pairs.
(3) Improve tools for data analysis.
(4) All of these.

20) Which is not related to minisatellites ?

(1) Non-coding sequencies
(2) High degree polymorphism
(3) 11-60 base pairs repeats
(4) Non-inheritable sequences

21) Which of the following statement is correct Regarding translation ?

(1) Presence of catalyst would enhance Rate of peptide bond formation.
(2) In Eukaryote 23S rRNA is used for peptide bond formation
(3) Ribosome slide over mRNA in 3'→5' direction
(4) During start of translation Small subunit of Ribosome Encounter a t-RNA at first.

22) Which of the following is used as a substrate by RNA polymerase and polymerises in a template depended fashion ?

(1) Ribonucleoside triphosphates
(2) deoxyribonucleoside triphosphates
(3) Ribonucleoside monophosphates
(4) deoxyribonucleoside monophosphates

23) During DNA Replication okazaki Fragment are used to Elongate :-

(1) The leading strand toward Replication fork
(2) The lagging strand toward Replication fork
(3) The leading strand away from Replication fork
(4) The lagging strand away From Replication Fork

24) Select correct statements about predation :
(a) Predators help to maintain species diversity in population.
(b) In a field experiment when all star fishes Pisaster were removed from enclosed intertidal area, more than 10 species of vertebrates became extinct within a year.
(c) Predator is too efficient and over exploits its prey
(d) Predator is prudent in nature
(e) Camauflage is method to avoid being easily detected by predator.

(1) All are correct
(2) Only a, b, c are correct
(3) Only a, d, e are correct
(4) Only d is correct

25) In a tropical rain forest three study sample areas are selected
In ***sample area1*** *:* Three bird species are present. Two species are represented by one individual each while the third species has four individuals. In ***sample area 2*** *:* This area has same 3 species each species represented by two individuals each. *In* ***sample are a 3*** *:* This area has one bird species in common with sample area 1 and 2 with 2 individuals, second species represented an insect with 2 individuals and third species represented a mammal with 2 individuals.
So species diversity is highest in which area?

(1) sample area 1

(2) sample area 2

(3) sample area 3

(4) Both the area 2 & 3

26) Given below are two statements :- one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) :- Totally unrelated species could also compete for the same resource.
Reason (R) :- In some shallow south American lakes, vising flamingoes and resident fishes compete for their common food, the zooplankton in the lake.
In the light of the above statements, choose the correct answer from the options given below :-

(1) Both A and R are true but R is not the correct explanation of A

(2) A is true but R is false

(3) A is false but R is true

(4) Both A and R are true and R is the correct explanation of A

27) Which of the following statements are **incorrect** ?
(A) Even in minimal communities, many interactive linkages exist, although all may not be readily apparent.
(B) Predation, parasitism and commensalism share a common characteristic - the interacting species live closely together.
(C) Biological control methods adopted in agricultural pest control are based on the ability of the prey to regulate predator population.
(D) Darwin was convinced that interspecific competition is a potent force in organic evolution.

(1) A, B and C

(2) Only B

(3) Only C

(4) A and C

28) Match **list-I** with **list-II** :-

	List-I (Interacting species)		**List-II (Name of interatction)**
A.	Abingdon tortoise and goats in galapagos island	I.	Commensalism
B.	Marine fish are infested with copepods	II.	Mutualism

C.	An orchid growing on a mango branch	III.	Competition
D.	Fungi and algae in lichens	IV.	Parasitism

Choose the **correct** answer from the options given below :

(1) A-III, B-IV, c-II, D-I

(2) A-III, B-IV, c-I, D-II

(3) A-IV, B-III, c-I, D-II

(4) A-IV, B-III, c-II, D-I

29) In a hypothetical population of 100 individual having r = 0.4/individual/yr. What will be the population size in 5 years (with e = 2.72) showing exponential rate of growth?

(1) 1218

(2) 739

(3) 2012

(4) 448

30)

Which among the following statement is correct?

(1) Bamboo and pelagic fishes breed only once in their lifetime.

(2) Oysters produce a large number of small sized off-springs.

(3) Organisms evolve towards the least efficient reproductive strategy.

(4) Birds produce a large number of large sized offsprings.

31) If Population of an insect species increases explosively beyond carrying capacity during rainy season and then disappears rapidly at the end of the season, it shows that :

(1) Food plants mature and die at the end of rainy season

(2) Population of predators increases enormously

(3) Population growth curve is J-shaped

(4) Population growth curve is S-shaped

32) Ecology explains us that :-

(1) How organism, while remaining as an individual interact with other organisms.

(2) How organism, interact with physical environment.

(3) As a group how organisms behave like organised population, community, ecosystem or even as the whole biosphere.

(4) All of these

33) How many listed factors can affect the rate of decomposition ?
Warm & moist environment, Anaerobiosis, Chemical composition of detritus, pH of soil

(1) 4

(2) 3

(3) 2

(4) 1

34)

Fill in the blank

The _____ productivity results in the accumulation of plant biomass, which serves as the food of herbivores and decomposers :-

(1) Gross primary

(2) Net primary

(3) Gross Secondary

(4) Net Secondary

35) Choose the **incorrect** statement :-

(1) The pyramid of biomass shows standing crop

(2) Pyramid of number shows biotic potential

(3) Pyramid of biomass is upright in aquatic ecosystem

(4) Pyramid of number in a tree ecosystem is inverted

36)

Which among the following statement is/are incorrect -
(a) Energy flow in the ecosystem is exempt from first law of thermodynamics.
(b) In a simple food chain the energy is lost in form of heat.
(c) Top consumers are ecologically weakest organisms of nature.

(1) b and c

(2) a and c

(3) a only

(4) c only

37) With respect to energy flow which of the following is correct.

(1) Out of the incident solar radiation less than 50% of it is PAR

(2) Plants capture only 2-10% of the PAR and this small amount of energy sustains the entire living world

(3) There is unidirectional flow of energy from the sun to producers and then to consumers

(4) All of these

38) How many statements are incorrect :-
(a) The number of trophic levels in the grazing food chain is restricted.
(b) Organism at each trophic level depend on those at the higher trophic level for their energy demands
(c) Each trophic level has a certain mass of living material at a particular time called as the standing

crop
(d) Decomposers has the largest population in grazing food chain

(1) Three

(2) Four

(3) Two

(4) One

39)

Given below are two statements; one is labelled as **Assertion (A)** and the other a labelled as **Reason (R).**
Assertion (A): A sparrow is a primary consumer when it eats seeds, fruits, peas, and a secondary consumer when it eats insects and worms.
Reason (R): A given species may occupy more than one trophic level in the same ecosystem at the same time.

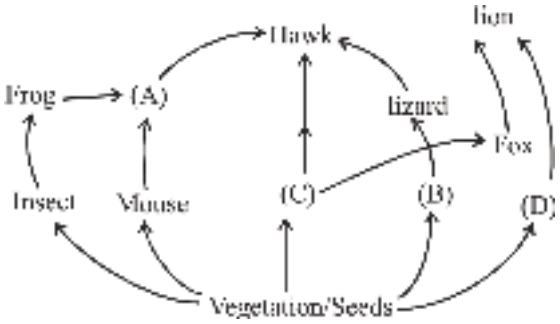
(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

40) Identify the likely organisms (A), (B), (C) and (D) in the food web shown below :-



Option:-

	A	B	C	D
(1)	Snake	Dog	Deer	Rabbit
(2)	Snake	Grasshopper	Rabbit	Deer
(3)	Rabbit	Grasshopper	Snake	Cat
(4)	Deer	Crow	Pigeon	Tortoise

(1) 1

(2) 2

(3) 3

(4) 4

41) Match the column I and II choose the correct combination

	Column-I		Column-II
(K)	Hotspots	(a)	34
(L)	National parks	(b)	90
(M)	Wildlife Sanctuaries	(c)	14
(N)	Biosphere Reserves	(d)	448

(1) K-a, L-b, M-c, N-d

(2) K-a, L-d, M-c, N-b

(3) K-a, L-c, M-b, N-d

(4) K-a, L-b, M-d, N-c

42) **Statement-I:** Government of India in 1980s has introduced the concept of joint forest management (JFM).
Statement-II: Cichlid fish introduced into Lake Victoria in east Africa led eventually to the extinction of more than 200 species of Nile perch in the lake.

(1) Statement I and II both are correct

(2) Statement I and II both are incorrect

(3) Only Statement I is correct

(4) Only Statement II is correct

43) Select the incorrect statement -

(1) Protection of hot spot can induce ongoing mass extinction by almost 30%

(2) India covers about 2.4% of world land area and shares 8.1% of total global diversity.

(3) Exploring molecular, genetic and species level for products of economic importance is called Bio-prospecting.

(4) The world summit on sustainable development was held in 2002.

44) Match column I with column II :-

Column-I		Column-II	
(a)	Over-exploitation	(i)	Amazon rain forest
(b)	Co-extinction	(ii)	*Lantana, Eicchornia*
(c)	Habitat loss	(iii)	Plant & Pollinator
(d)	Alien species invasion	(iv)	Stellers sea cow, Passenger pigeon

Options :-

(1) a – iv, b – ii, c – i, d – iii

(2) a – ii, b – iv, c – i, d – iii

(3) a – iv, b – iii, c – i, d – ii

(4) a – i, b – ii, c – iii, d – iv

45) In "rivet popper hypothesis" (Paul Ehrlich) rivet indicates :-

(1) Ecosystem
(2) Biosphere
(3) Community
(4) Species

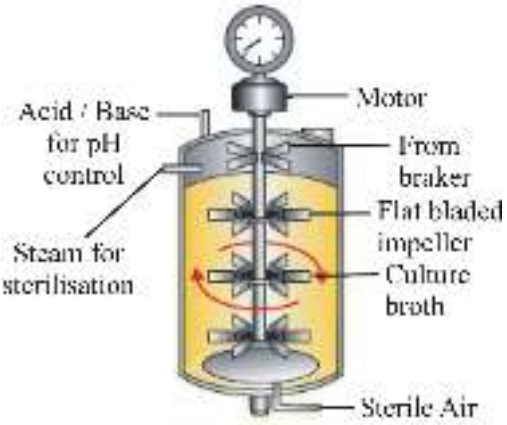
46) Bt-cotton has which of the following special features?

(1) This plant is completely resistant to all insects.
(2) It requires less fertilizers.
(3) It's leaf is resistant to pest but boll is destroyed by bollworms.
(4) This plant is resistant to certain insects.

47) Which of the following is not a part of biotechnology ?

(1) Gene therapy
(2) *In vitro* fertilization leading to a test tube baby
(3) Developing DNA vaccine
(4) Biomagnification of pesticide

48) Given below in the figure of one of the most commonly used bioreactor.



In this diagram flat bladed impeller helps in :-

(1) Maintaining the pH of medium
(2) Stirring the liquid medium
(3) Regulating temperature of medium
(4) Both (1) & (3)

49) Read the following four statements (A-D) about certain mistakes in two of them.
(A The first transgenic buffalo, "Rosie" produced milk which was human alpha-lactalbumin enriched.
(B) Restriction enzymes are used in isolation of DNA from other macro molecules.
(C) Downstream processing is one of the steps of r-DNA technology.
(D) Disarmed pathogen vectors are also used in transfer of r-DNA into the host. Which are the two statements having mistakes ?

(1) Statements (A) and (B)

(2) Statements (B) and (C)

(3) Statements (C) and (D)

(4) Statements (A) and (C)

50) Match the Column-I with Column-II :-

Column - I		Column - II	
(i)	Primers	(A)	PCR
(ii)	Separation & purification of products	(B)	Chilled C_2H_5OH
(iii)	Precipitation of DNA	(C)	Gene gun
(iv)	Gene transfer	(D)	Downstream processing

(1) (i) - A, (ii) - D, (iii) - B, (iv) - C

(2) (i) - B, (ii) - A, (iii) - D, (iv) - C

(3) (i) - D, (ii) - A, (iii) - C, (iv) - B

(4) (i) - A, (ii) - B, (iii) - D, (iv) - C

51) Find the correct match for PCR :-

(1) Denaturation - Use of taq polymerase

(2) Annealing - Attachment of primers

(3) Extension - At a temperature of 94°C

(4) Denaturation-Breakdown of phosphodiester bonds

52)

Match the following.

	Column-I		Column-II
(A)	Stanley Cohen and Herbert Boyer	(I)	Separation of DNA fragments
(B)	Hind II	(II)	Construction of first recombinant DNA
(C)	Gel electrophoresis	(III)	First restriction endonuclease
(D)	Bam HI site	(IV)	At tetracycline resistance gene in vector pBR322

(1) A-II, B-III, C-IV, D-I

(2) A-II, B-III, C-I, D-IV

(3) A-II, B-IV, C-I, D-III

(4) A-IV, B-III, C-II, D-I

53) Which of the following statement is not correct ?

(1) Each restriction endonuclease recognises a specific palindromic nucleotide sequences in the DNA.

(2) The separated bands of DNA are cut out from the agarose gel and extracted from the gel piece,this is known as elution.

(3) Any protein encoding gene is expressed in a heterologous host then protein is called a recombinant protein.

(4) In Gel electrophoresis the DNA fragements loaded towards positive electrode.

54)

Consider the following statement and select the **correct** option :-
(A) Any protein encoding gene is expressed in a heterologous host, is called a recombinant protein
(B) Stirred tank reactor facilitates even mixing and oxygen availability throughout the bioreactor.
(C) The downstream processing and quality control testing is similar to all products.

Options :

(1) A and C are correct while B is wrong.

(2) A and B are correct while C is wrong.

(3) C and B are correct while A is wrong.

(4) Only A is correct while B and C are wrong.

55) Which of these are applications of biotechnology?
a. Therapeutics and diagnostics
b. Genetically modified crops
c. Processed food
d. Energy production
e. Waste treatment & bioremediation

(1) b, c, d & e

(2) a, b, c & e

(3) a, b, d & e

(4) a, b, c, d & e

56) Bt cotton plants are :-

(1) Hybrid plants resistant to cotton bollworms.

(2) Hybrid plants resistant to corn borer.

(3) Transgenic plants resistant to cotton bollworms.

(4) Transgenic plants resistant to *Meloidegyne incognitia*.

57) Which of the following is wrongly matched in the given table ?

(i)	Golden rice	Transgenic rice	Vitamin A enriched
(ii)	Flavr savr	Transgenic tobacco	Delayed ripening
(iii)	Bt-cotton	Transgenic cotton	Bacterial and viral resistance

(iv)	Bacillus	Insecticidal protein	RNA interference

(1) i and ii

(2) ii, iii and iv

(3) ii and iii

(4) Only iii

58) 'Rosie' a transgenic cow is known to produce a type of milk, which has all the following characteristics except :-

(1) Protein content of 2.4 g / litre.

(2) Has human α-lactalbumin protein.

(3) Can be used for treatment of cystic fibrosis.

(4) More balanced diet than normal cow milk for babies.

59) Which of the following statements is incorrect about gene therapy (1990) in ADA deficiency?

(1) Lymphocytes from patient's blood are taken out and cultured.

(2) A functional ADA cDNA is introduced into these lymphocytes.

(3) Genetically engineered lymphocytes are introduced into the body of patient.

(4) Patient does not require periodic infusion of genetically engineered lymphocytes.

60) Match column-I with column-II and select the correct answer from the codes given below.

Column-I		**Column-II**	
(A)	α-1 -antitrypsin	(i)	AIDS
(B)	GEAC	(ii)	Gene therapy
(C)	ELISA	(iii)	Emphysema
(D)	Retroviral vector	(iv)	Biosafety

(1) A-(i), B-(iii), C-(ii), D-(iv)

(2) A-(iii), B-(iv), C-(i), D-(ii)

(3) A-(i), B-(ii), C-(iii), D-(iv)

(4) A-(iii), B-(i), C-(ii), D-(iv)

61) A genetically engineered micro-organism used successfully in bioremediation of oil spills is a species of :-

(1) *Pseudomonas*

(2) *Trichoderma*

(3) *Xanthomonas*

(4) *Bacillus*

62) Gene therapy performed for correction of disorder :-

(1) is always a temporary cure.

(2) could be permanent if gene is introduced in the early embryonic stages.

(3) has not been tried for humans.

(4) involves removal of functional genes & introduce of Non functional genes.

63)

How many of the following statements are incorrect:
I. Gene of *Agrobacterium* is used to produce humulin.
II. Indian government has set up GEAC.
III. α-1-antitrypsin is used to treat PKU.
IV. ELISA serve the purpose of early diagnosis.
V. *Flav-savr* is variety of pomato.

(1) One

(2) Two

(3) Three

(4) Four

64) Full potential of Penicillin as an antibiotic was established by :-

(1) Chain and Fleming

(2) Chain and Florey

(3) Chain, Florey and Fleming

(4) Florey and Fleming

65) Name the blank space A, B, C and D given in the following table -

Microbes name	**Commercial product**
A	Lactic acid
B	Cyclosporin- A
C	Penicillin
D	Citric acid

(1) A - *Lactobacillus* B-*Trichoderma* (fungi)
C - Penicillin D - Statin
(2) A - *Trichoderma* B - Statin
C - *Lactobacillus* D - Penicillin
(3) A - *Lactobacillus* B - *Trichoderma* (fungi) C - *Penicillium notatum* D - *Aspergillus* (4) A - *Lactobacillus* B - *Trichoderma*
C - Penicillin D - *Streptococcus*

(1) 1

(2) 2

(3) 3

(4) 4

66)

Choose the **correct** statement :-

(1) *Aspergillus niger* is bacterium which is used for obtaining acetic acid & citric acid.

(2) Streptokinase is used as a clot buster.

(3) *Monascus purpureus* is responsible for production of large holes in Swiss cheese.

(4) Toddy is manufactured by *Lactobacillus.*

67) Which of the following pairs is **wrongly** matched ?

(1)	Lactic acid Bacteria	-	Partially digest the milk proteins
(2)	Cheese	-	Oldest food items in which microbes were used
(3)	Streptokinase	-	Used as an immunosuppressive agent in organ transplant patients
(4)	Lady bird	-	Useful to get rid of Aphids

(1) 1

(2) 2

(3) 3

(4) 4

68)

Match the following :

	Column 'A'		**Column 'B'**
A.	1gA	P.	Can cross Placenta
B.	1gG	Q.	Found in colostrum
C.	1gM	R.	First Antibody produced
D.	Antibody Structure	S.	H_2L_2

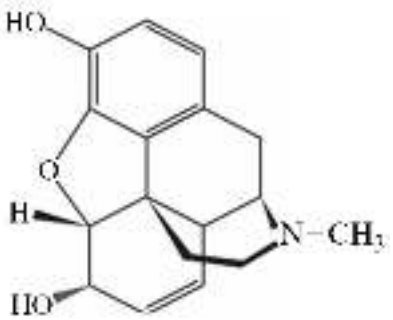
(1) A-Q, B-P, C-R, D-S

(2) A-P, B-Q, C-R, D-S

(3) A-Q, B-P, C-S, D-R

(4) A-Q, B-R, C-P, D-S

69) Choose the incorrect statement w.r.t. the given chemical structure



(1) Its diacetylated form is termed as smack.

(2) It is extracted from the latex of poppy plant.

(3) It is a depressant and slows down body functions.

(4) Its receptors are present principally in our CNS and cardio vascular system.

70) Which of the following statement is incorrect?

(1) Heat and moisture helps the fungi in their growth.

(2) Hepatitis-B vaccine is an example of second generation vaccine.

(3) Typhoid Mary was a cook by profession and was a typhoid carrier.

(4) Male *Aedes* mosquito is a vector for disease filariasis.

71) Choose the correct statement of events in life cycle of malarial parasite.

(1) Parasites reproduce asexually in the RBCs of human, bursting the RBCs and causing cycles of fever.

(2) Sexual stages (gametocytes) develop in the intestine of humans.

(3) Fertilisation and development takes place in the salivary gland of female Anopheles mosquito.

(4) Female Anopheles mosquito takes up gametes with blood meal.

72) Choose the incorrect match–

(1)	Malignant malaria	Plasmodium falciparum
(2)	Plague	Viral disease
(3)	Trichophyton	Ringworm
(4)	Common cold	Rhino virus

(1) 1

(2) 2

(3) 3

(4) 4

73) An antibody is _____

(1) The molecule that binds specifically as antigen

(2) WBC which invades bacteria

(3) Secretion of mammalian RBC

(4) The cellular component of blood

74) Many diseases can be diagnosed by observing the symptoms in the patient. Which group of symptoms are indicative of pneumonia?

(1) Difficulty in respiration, fever, chills, cough, headache

(2) Constipation, abdominal pain, cramps, blood clots

(3) Nasal congestion and discharge, cough, sorethroat, headache

(4) High fever, weakness, stomach pain, loss of appetite and constipation

75) The process of spreading the cancerous cells to distant site is known as :-

(1) Hyperstasis
(2) Metastasis
(3) Parastasis
(4) Parasitesis

76) Which one of the following statements is correct about T-lymphocytes in mammals?

(1) These are produced in thyroid.
(2) There are only three main types - cytotoxic T-cells, helper T-cells and suppressor T-cells.
(3) These are originated in lymphoid tissues.
(4) They scavenge damaged cells and cellular debris.

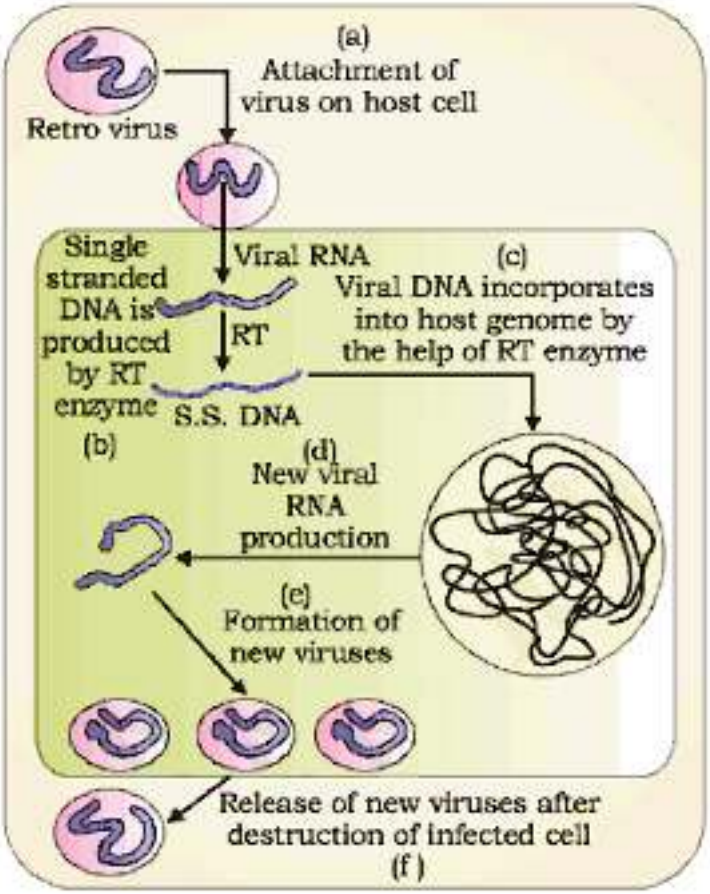
77) Which of the following is used in the production of the recombinant Hepatitis B vaccine?

(1) Yeast
(2) Microsporum
(3) Ecoli
(4) Streptococcus

78) Given below are two statements: One is labelled as **Assertion A** and the other is labelled as **Reason R**.
Assertion A: Some human organs like liver, kidney fail to function normally and transplantation is the only remedy to enable the patient to live a normal life.
Reason R : Tissue matching and blood group matching are essential before undertaking any transplant and patient has to take immuno-suppresant thereafter.
In the light of the above statements, **choose** the **correct** answer from the options given below :

(1) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.
(2) **A** is true but **R** is false.
(3) **A** is false but **R** is true.
(4) Both **A** and **R** are true and **R** is the correct explanation of **A**.

79) The following diagram shows the replication of retrovirus in host cell. Select the option that includes correctly marked steps of this process:



(1) d,b,f

(2) b,e,f

(3) a,d,e

(4) a, c, f

80)

Match the hominids with their correct cranial capacity :

	Column I		**Column II**
(a)	Homo habilis	(i)	900 cc
(b)	Homo neanderthalensis	(ii)	1350 cc
(c)	Homo erectus	(iii)	650-800 cc
(d)	Homo sapiens	(iv)	1400 cc

Select the correct option.

Options:	(a)	(b)	(c)	(d)
1.	(iv)	(iii)	(i)	(ii)
2.	(iv)	(i)	(iv)	(ii)
3.	(iii)	(ii)	(i)	(iv)
4.	(iii)	(iv)	(i)	(ii)

(1) 1
(2) 2
(3) 3
(4) 4

81) The land reptile which went back into water to evolve into fish like reptiles–

(1) Around 250 mya e.g. Ichthyosaurs
(2) Around 300 mya e.g. Therapsids
(3) Around 200 mya e.g. Ichthyosaurs
(4) Around 150 mya e.g. Therapsids

82) Which of the following provides evidence for evolution from a common ancestor?

(1) Comparative anatomy
(2) Embryology
(3) Fossil records
(4) All of the above

83) Which of the following structures are different in fundamental structure but evolving for the same function as a result of convergent evolution :-

(1) Vertebrates brain
(2) Flippers of penguin and dolphins
(3) Heart of pigeon and rabbit
(4) Mouth parts of insects

84) Following figure represent-



(1) Variety of eyes of finches that lamarck found in galapagos island.
(2) Variety of feathers of finches that haldane found in England.
(3) Variety of beaks of finches that darwin found in galapagos island.
(4) Variety of head of finches that de Vries found in Holand.

85) Match the evolution concepts given in column-I with column-II and select the correct answer.

Column - I		**Column - II**	
A.	Mutation	I.	Change in population's allele frequencies due to chance alone.

B.	Gene flow	II.	Differences in survival and reproduction among variant individuals.
C.	Natural selection	III.	Immigration, emigration change allele frequencies.
D.	Genetic drift	IV.	Source of new alleles.

(1) A – I, B – II, C – III, D – IV

(2) A – IV, B – II, C – III, D – I

(3) A – III, B – I, C – IV, D – II

(4) A – IV, B – III, C – II, D – I

86) The man like primates walked in eastern Africa around 3-4 mya. This belief was supported by

(1) Hominid features of bones discovered in Ethiopia and Tanzania.

(2) Hominid features of bones discovered in whole Africa.

(3) Hominid features of bones discovered in Peru.

(4) Hominid features of bones discovered in Australia.

87) Given below are two statements :
Statement-I : Fossils are remains of hard parts of life forms found in rocks.
Statement-II : In 1938, a fish caught in South Africa happened to be a coelacanth which was thought to be extinct.

(1) Statement-I and Statement-II both are correct.

(2) Statement-I is incorrect but Statement-II is correct.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I and Statement-II both are incorrect.

88) Given below are two statement.
Statement-I :- Fossils are remain of hard parts of life forms found in rocks.
Statement-II :- The geological history of earth closely correlates with the biological history of earth.

(1) Statement-I is true but statement-II is false.

(2) Statement-I is false but statement-II is true.

(3) Both statement-I and statement-II are true.

(4) Both statement-I and statement-II are false.

89) Which of the following is not correct about genetic drift ?

(1) Change in gene frequency in a small population occurring by chance

(2) It does not upsets Hardy–Weinberg equilibrium

(3) It fixes or eliminates genes in the gene pool irrespective of its adaptive value

(4) Sometimes the change in allele frequency is so different that the original drifted population becomes founders

90) Before industrialization, thick growth of almost white coloured lichen covered the trees-in that background the white winged moth survived but the dark-coloured moth were picked out by predators.
In respected of this paragraph the event should be:-

(1) Stabilizing selection

(2) Directional selection

(3) Disruptive selection

(4) Artificial selection

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	3	2	4	1	3	3	2	1	2	1	4	4	4	1	4	4	1	2	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	2	2	3	3	1	2	4	3	3	3	1	2	1	3	2	3	3	2	1
Q.	41	42	43	44	45															
A.	2	1	4	3	1															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	4	4	1	3	3	4	2	3	2	3	2	3	3	2	1	4	3	2	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	2	4	4	1	4	3	1	2	2	2	1	1	2	2	4	1	3	2	1
Q.	86	87	88	89	90															
A.	4	3	2	4	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	3	4	4	1	1	4	3	2	3	2	4	1	1	4	1	1	3	4	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	1	4	3	3	4	3	2	2	2	3	4	1	2	3	3	4	3	4	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	3	1	3	4	4	4	2	1	1	2	2	4	2	4	3	2	3	4	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	1	2	3	2	3	2	3	1	4	4	1	2	1	1	2	2	1	1	3	4
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	4	2	3	4	1	1	3	2	2										

SOLUTIONS

PHYSICS

1)

Magnetic force : $\vec{F}_B = q(\vec{v} \times \vec{B})$

Electric force : $\vec{F}_E = q\vec{E}$

Magnetic force $\vec{F}_B$ acts Perpendicular to direction of $\vec{v}$ so, it can only charge direction of $\vec{v}$.
Statement I is correct. If $\vec{E}$ is not parallel to $\vec{v}$ it, can change the direction of velocity and hence deflect the charge. Therefore, an electric field can both change speed and deflect the charge.
Statement-II is incorrect.

2) (a) $B_C = \frac{\mu_0 i}{4r}(-\hat{k})$

(b) $B_C = \frac{\mu_0 i}{4r}(+\hat{k})$

(c) $B_C = \frac{\mu_0 i}{2r}\hat{k} + \frac{\mu_0(2i)}{4(2r)}(-\hat{k})$

$= \frac{\mu_0 i}{4r}\hat{k}$

(d) $B_C = \frac{\mu_0(2\hat{i})}{4(2r)}(+\hat{k}) + \frac{\mu_0 i}{2r}(-\hat{k})$

$= \frac{\mu_0 i}{4r}\hat{k} - \frac{\mu_0 i}{2r}\hat{k}$

$= = \frac{\mu_0 i}{4r}(-\hat{k})$

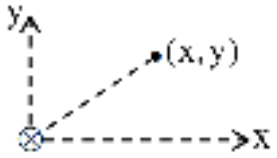
3)

Magnetic field inside a solenoid is $B = \mu_0 nI$ where n is total turns per unit length.
Total turn $N = 5 \times 400 = 2000$
$L = 80\ cm = 0.8\ m$
$I = 8.0\ A$
$n = \frac{2000}{0.8} = 2500\ turns/m$
$B = 4\pi \times 10^{-7} \times 2500 \times 8$
$B = 4\pi \times 10^{-7} \times 20000$
$B = 8\pi \times 10^{-3}\ T$

4) $f = \frac{\mu_0 i_1 i_2}{2\pi d} = \frac{\mu_0 i^2}{2\pi d}$

$f \propto \frac{i^2}{d}$

5)



$\vec{B} = B\cos\theta\hat{i} - B\sin\theta\hat{j}$

$\vec{B} = \dfrac{\mu_0 i(y\hat{i} - x\hat{j})}{2\pi(x^2 + y^2)}$

6) Resonant frequency, $\omega_0 = \dfrac{1}{\sqrt{LC}} = 10^4$ rad/sec

$\omega' = .4 \times 10^4 = 4000$ rad/sec

$i_0 = \dfrac{V_0}{\sqrt{R^2 + (X'_C - X'_L)^2}} = 238$ mA

7) $\cos\phi = \dfrac{R}{Z}$

$0.8 = \dfrac{R}{Z}$

$Z = \dfrac{R}{0.8}$

$\sqrt{R^2 + X_L^2} = 1.25\ R$

$X_L = 0.75\ R$

$P = \left(\dfrac{V}{Z}\right)^2 R$

$400 = \left(\dfrac{250}{Z}\right)^2 R = \left(\dfrac{250}{1.25R}\right)^2 R$

$R = 100\ \Omega$

For $\cos\phi = 1$

$X_L = X_C$

$0.75\ R = \dfrac{1}{2\pi fC}$

$C = \dfrac{1}{2\pi f \times 0.75 \times 100}$

8)

velocity of EM wave is given by $\dfrac{1}{\sqrt{\mu \in}}$

velocity in air $= \dfrac{\omega}{k} = C$

velocity in medium $= \dfrac{C}{2}$

$\dfrac{\frac{1}{\sqrt{\in_{r_1}}}}{\frac{1}{\sqrt{\in_{r_2}}}} = \dfrac{C}{\left(\frac{C}{2}\right)} = 2 \Rightarrow \dfrac{\in_{r_1}}{\in_{r_2}} = \dfrac{1}{4}$

9) $\phi_i = NBA$; $\phi_f = NBA\cos 60° = \frac{NBA}{2}$

$\Rightarrow \Delta\phi = \phi_f - \phi_i = \frac{-NBA}{2}$

$\Rightarrow e_{ind} = \frac{-\Delta\phi}{\Delta t} = 2\pi \times 10^{-2} V$

$\Rightarrow q = \frac{-\Delta\phi}{R} = \frac{2\pi \times 10^{-3}}{4} = \frac{\pi}{2} \times 10^{-3} C$

10) Current in B_1 and current in B_2 will slowly tend to zero, because of magnetic energy store in inductor.

11)

Since induce current will be in anticlock wise direction.

12) $\frac{d^2v}{dt^2} + \frac{(B\ell)^2}{mL} v = 0$

$\Downarrow$

Simple harmonic equation

$\Downarrow$

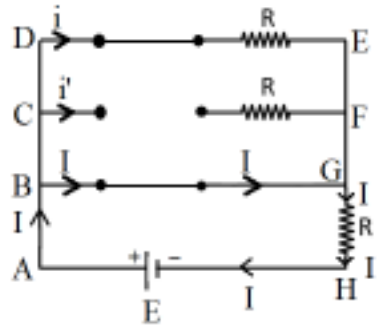
Displacement vs time graph ⇒ Oscillatory

13) $\lambda = \frac{12400}{E} = \frac{12400}{1.5}$

λ = 8266 Å (IR rays)

14)

Equivalent circuit



i' = 0 (as the diode D_2 is in Reverse Bias)

And, the diodes D_1 and D_3 are in Forward Bias and arm BG is short circuited.

So, i = 0

$I = \frac{E}{R}$

15) Y = A . B + (A + B). A

= A.B + A. A + A.B

= A.B + A + A.B

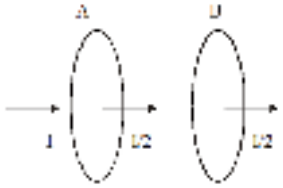
= A (B + 1 + B)
= A

16) Independent sources are not coherent.

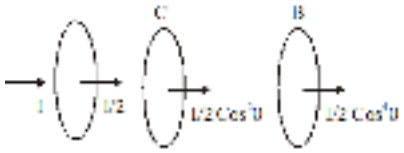
17) Shift of central maxima = nλ = (μ – 1)t
$40 \times 600 \times 10^{-9} = (2 - 1)\, t$
$t = 24000 \times 10^{-9}$
$= 24 \times 10^{-6}$
$\boxed{t = 24\,\mu m}$

18) $\beta = \frac{\lambda . D}{d}$
$\lambda_R > \lambda_V$
$\beta_R = \frac{\lambda_R D}{d}$ and $\beta_V = \frac{\lambda_V D}{d}$
$\beta_R > \beta_V$
Fringe pattern will shrink.

19) Axis of transmission of A & B are parallel.



Now,



$\frac{I}{2}\cos^4\theta = \frac{I}{8} \Rightarrow \cos^4\theta = \frac{1}{4}$
$\cos\theta = \frac{1}{\sqrt{2}}$
$\theta = 45°$

20)

When light hits a surface, it intracts with the matter. This intraction results in a force per unit area, known as radiation pressure.
According to the quantum theory of light, photons (the particles of light) Carry momentum.
The momentum (P) of a photon is related to its energy (E) by the formula $P = \frac{E}{C}$.

21) $F = \frac{2 \times I \times A}{C}$

$$= \frac{2 \times 50 \times 10^{-3} \times 9 \times 10^{-6}}{3 \times 10^{8}}$$
$= 3 \times 10^{-15}$ N

22) $\frac{\phi_{0A}}{\phi_{0B}} = \frac{1}{2} \Rightarrow \phi_{0B} = 2\phi_{0A}$ .... (1)
$K_A = hf - \phi_{0A} \Rightarrow hf = K_A + \phi_{0A}$ .....(2)
$2hf = K_B + 2\phi_{0A}$ .....(3)
$$\frac{1}{2} = \frac{K_A + \phi_{0A}}{K_B + 2\phi_{0A}}$$
$$\frac{K_A}{K_B} = \frac{1}{2}$$

23) $$\lambda = \frac{h}{\sqrt{2mE}}$$
$$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{E_2}{E_1}}$$
$$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{169}{100}} = \frac{13}{10}$$
$$\frac{\Delta\lambda}{\lambda}\% = \left[\frac{10}{13} - 1\right] \times 100\%$$
$$= -\frac{300}{13} = -23\%$$
wavelength decreases by 23%

24) $$\lambda = \frac{h}{\sqrt{2mqv}}$$
$$\frac{v_1}{v_2} = \frac{m_2}{m_1} \times \frac{q_2}{q_1}$$
$$\frac{100}{v_2} = \frac{7m}{m} \times \frac{2e}{e}$$
$v_2 = 7.14$ volt

25) Z – 2 × 2 + 2 × 1 – 4 × 1 ⇒ Z – 6
and A – 2 × 4 ⇒ A – 8
Number of neutrons = A – 8 – [Z–6] = A – Z – 2
$$\frac{N}{Z} = \frac{A - Z - 2}{Z - 6}$$

26) COLM
$(A-4)V_N = 4V$
$$V_N = \frac{4V}{(210-4)} = \frac{4V}{206}$$

27) $$\frac{1}{f_1} = (1.2 - 1)\left[\frac{1}{\infty} - \frac{1}{(-14)}\right] = \frac{0.2}{14}$$

$$\frac{1}{f_2} = (1.5-1)\left[\frac{1}{14} - \frac{1}{\infty}\right] = \frac{0.5}{14}$$

$$\frac{1}{f_{eq}} = \frac{0.7}{14} = \frac{1}{20}$$

$$\frac{1}{v} - \frac{1}{(-40)} = \frac{1}{20} \Rightarrow \frac{1}{V} = \frac{1}{20} - \frac{1}{40} = \frac{1}{40}$$

$$\boxed{V = 40\,cm}$$

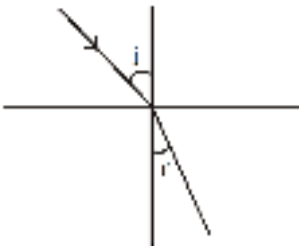
28)

Length of image $= \left(\frac{f}{f+u}\right)^2 h_0$

Length of image $= \left(\frac{10}{10-15}\right)^2 \times 1mm = 4mm$

29) M = 1 + D/f
Given D = 25 cm and f = 5 cm
M = 1 + 25/5
M = 1 + 5
M = 6

30) Position of object Position of image
–∞ –C between C and F
C – F between –∞ and C
F – P between P and +∞
So for any position of object image can't be between f and P.

31) 

1 sin i = μ sin r
i × sin 60 = $\sqrt{3}$ sin r

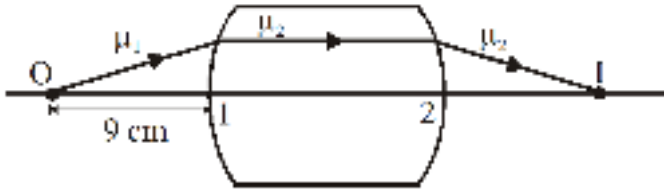
$$\sin r = \frac{1}{2}$$

r = 30
sin i = μ sin r

$$\cos i\frac{di}{dt} = \mu \cos r\frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{1}{\mu}\left(\frac{\cos i}{\cos r}\right)\frac{di}{dt}$$

$$= \frac{1}{\sqrt{3}}\frac{\cos 60}{\cos 30} \times 3 = 1 \text{ rad/sec}$$



32)

At surface 1 : $u_1 = -9$, $\mu_1=1$, $\mu_2=2$, $R = +9$;

$$\frac{2}{v_1} - \frac{1}{-9} = \frac{2-1}{+9} \Rightarrow \boxed{v_1 = \infty}$$

At surface 2 : $u_2 = \infty$; $\mu_1 = 2$, $\mu_2 = 1$, $R = -9$;

$$\frac{1}{v} - \frac{2}{\infty} = \frac{1-2}{-9} \Rightarrow \boxed{v = 9\ cm}$$

33) If we complete the trapezium it becomes an equilateral triangle.

$\therefore A = 60°$

$$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\frac{A}{2}}$$

$$\sqrt{2} = \frac{\sin\left(\frac{60+\delta_m}{2}\right)}{\sin\frac{60}{2}}$$



$$\frac{60+\delta_m}{2} = 45^\circ$$

$$\boxed{\delta_m = 30^\circ}$$

34) Shift due to glass slab :

$$\Delta = t\left[1 - \frac{1}{\mu}\right] \text{ away}$$

35) He should direct beam of light at air angle to vertical which is slightly less than critical angle. So that beam after refraction goes gazing the sea surface.

36) For air $\frac{1}{f} = \left(\frac{\mu_g}{\mu_a} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ ... (i)

in liquid $\frac{1}{f'} = \left(\frac{\mu_g}{\mu_\ell} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ ... (ii)

eq. (i) ÷ (ii)

$$\frac{f'}{f} = \frac{\left(\frac{3}{2}-1\right)}{\left(\frac{3/2}{3}-1\right)} = \frac{1/2}{-1/2}$$

f' = –f = –10 cm diverging lense.

37)

Diode → Two terminal device which conduct only in one direction.
Capacitor → Two terminal device conduct AC but not conduct DC. behave same in both direction.
LED → Diode which emit light.
Resistor → Two terminal device, conduct current in both direction with same steady rate.

38)

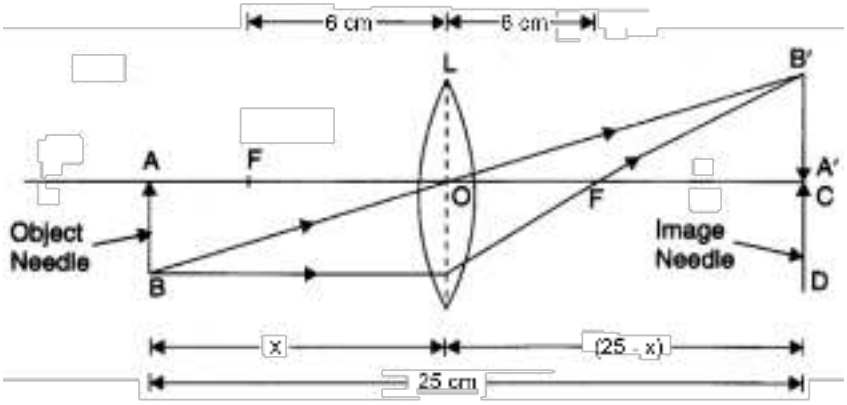
Reverse current in diode at breakdown is of the order of mA to Ampere.

39)

Least count of TM = $\left(\frac{1}{50}\right)\left(\frac{1}{20}\right)$ cm = 0.001 cm.
(1) Reading of microscope without slab (R_1)
= 3.20 + (4 × 0.001) = 3.204 cm
(2) Reading of microscope with slab when focussed on mark (R_2)
= 4.60 + (47 × 0.001) = 4.647 cm
(3) Reading of microscope on surface with Lycopodium pow. (R_3)
= 8.25 + (4 × 0.001) = 8.254,

$$\text{R.I.} = \frac{8.254-3.204}{8.254-4.647} = \frac{5.05}{3.607} = 1.4$$

40)



For the shown convex lens :
Object distance from lens ⇒ u = – x cm
Image distance from lens ⇒ v = + (25 – x) cm
Focal length ⇒ f = + 6 cm
Using Lens Formula :

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{(+6)} = \frac{1}{+(25-x)} - \frac{1}{(-x)}$$

$$\frac{1}{6} = \frac{1}{25-x} + \frac{1}{x}$$

$$\frac{1}{6} = \frac{(x) + (25 - x)}{(25 - x)(x)}$$

$$\frac{1}{6} = \frac{25}{25x - x^2}$$

$25x - x^2 = 150$

$x^2 - 25x + 150 = 0$

$x^2 - 15x - 10x + 150 = 0$

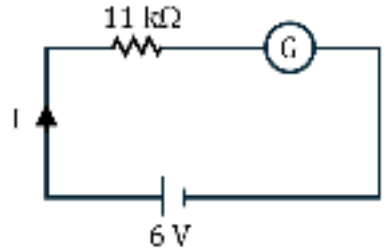
$x(x - 15) - 10(x - 15) = 0$

$(x - 15)(x - 10) = 0$

$x = 15$ or $x = 10$

So, the possible value (s) of object needle distance from the lens are 15 cm & 10 cm.

41)

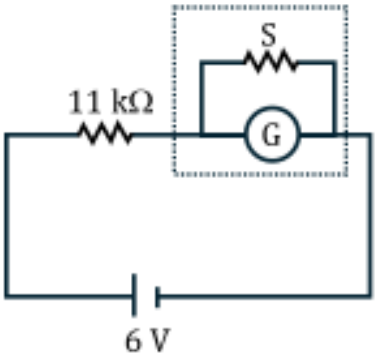


$I = 9 \times 60\ \mu A = 540 \times 10^{-6}\ A$

$$\frac{6}{11000 + G} = 540 \times 10^{-6}$$

$$G = \frac{1000}{9}$$

If shunt is applied



From the standard relation from half deflection method.

$$G = \frac{RS}{R - S} \Rightarrow \frac{1000}{9} = \frac{11000\,S}{11000 - S}$$

$S = 110\ \Omega$

42)

Magnetic field on axis:

$$B = \frac{\mu_0 I R^2}{2(R^2 + z^2)^{3/2}}$$

Radius, $R = 3\ cm = 0.03\ m$

Axial distance, $z = 4\ cm = 0.04\ m$

$R^2 = 9 \times 10^{-4}, \quad z^2 = 16 \times 10^{-4}$

$R^2 + z^2 = 25 \times 10^{-4}, \quad (R^2 + z^2)^{3/2} = (0.05)^3 = 1.25 \times 10^{-4}$

Substitute values:

$$B = \frac{4\pi \times 10^{-7} \times 2.5 \times 9 \times 10^{-4}}{2 \times 1.25 \times 10^{-4}}$$

$B = 36\pi \times 10^{-7}\ T$

Using right-hand rule, direction is along $+\hat{k}$.

43) Magnetic field cannot change speed and since magnetic field is along z axis, velocity cannot be in z axis at any instant i.e.. it cannot have $\hat{k}$ component.

44) Here, L = 2H, C = 32μF = 32 × 10^{-6} F, R = 10Ω

 Resonance frequency,

$$\omega_r = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{2 \times 32 \times 10^{-6}}} = \frac{10^3}{8}\ rad\,s^{-1}$$

Quality factor = $\frac{\omega_r L}{R} = \frac{10^3 \times 2}{8 \times 10} = 25$

45)

E_0 = 50 N/C

Cross-sectional area A = 10 cm^2 = 1 × 10^{-3} m^2

Length L = 50 cm = 0.5 m

$$E = \left[\frac{1}{2}\in_0 E_0^2\right] \times [Volume]$$

$$= \frac{1}{2} \times 8.85 \times 10^{-12} \times (50)^2 \times 50 \times 10 \times 10^{-6}$$

$$= 5.5 \times 10^{-12}\ J$$

CHEMISTRY

46)

(a) Fe^{+2} with NH_3 work as WFL so 4 unpaired e^-

(b) Co^{+3} with H_2O work as SFL so 0 unpaired e^-

(c) $[Ni(H_2O)_6]^{+2}$ 2 unpaired e^-

(d) $[Cr(CN)_6]^{-3}$ 3 unpaired e^-

47) $N(SiH_3)$ and boric acid show back bonding

48)

S1 → Pb^{+2} > Sn^{+2} (stability)

S2 → BiF_5 exist
S3 → Na_2HPO_2 does not exist

49)

	Hybridisation		**Unpaired e^-**
$[MnCl_6]^{-3}$	sp^3d^2	Outer orbital	4
$[FeF_6]^{-3}$	sp^3d^2	Outer orbital	5
$[CoF_6]^{-3}$	sp^3d^2	Outer orbital	4
$[Co(C_2O_4)_3]^{-3}$	d^2sp^3	Inner orbital	0

50) d^4 to d^7 strong field are more stable.

51) Ion-dipole force present in complex A.

52) (i) → Blue
(ii) → Violet / Purple
(iii) → Brown ring
(iv) → Crimson red

53) Scandium show only +3 oxidation state

54) Group - I → Pb^{+2}
Group - III → Fe^{+3}
Group - IV → Ni^{+2}
Group - VI → Mg^{+2}

55) $K_2Cr_2O_7 + H_2O_2 \rightarrow CrO_5$ (Blue)

56) $Fe^{+2} \xrightarrow{\text{Conc. } HNO_3} \text{Boil} \xrightarrow{\text{then}} \text{Cool} \rightarrow Fe^{+3}$
(Group III)

57) Na = Golden yellow
K = Violet
Ca = Brick red
Ba = Apple green

58) Nitrogen doesn't form pπ-dπ bond due to absence of vacant d-orbital.

59) Carbon family EO and EO_2 types oxides.

60) I_2O_5 can use to estimation of CO.

61) Its NCERT explaination

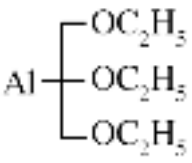
62) Highest density → Ir ≥ Os
Lowest density → Sc

63) Tb = $4f^9\ 6s^2$
Yb = $4f^{14}\ 6s^2$

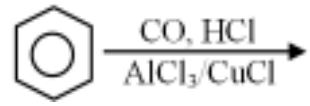
64) Cu^{+2} + excess I^- → Cu_2I_2 + I_2 $\xrightarrow{Na_2S_2O_3 \text{ solution}}$ NaI + $Na_2S_4O_6$

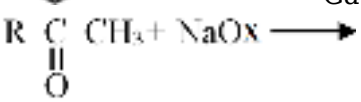
65) $CoCl_3.5NH_3$ → $[Co(NH_3)_5Cl]Cl_2$
0.1 mole
$\xrightarrow{\text{excess } AgNO_3}$ 0.2 moles AgCl
So compound will be $[Co(NH_3)_5Cl]Cl_2$
1 : 2 electrolytes

66) KCl / KF / KBr + I_2 → No reaction

67) 

68)

(a)  Gattermann-Koch Reaction

(b)  Haloform reaction

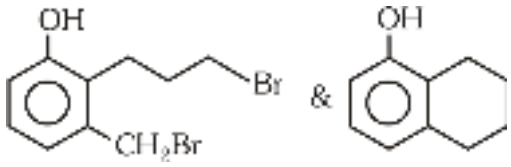
(c) R–CH_2–OH + R'COOH $\xrightarrow{\text{Conc. } H_2SO_4}$ Esterification

(d) $RO^{\ominus}\ Na^{\oplus}$ + R – X → Williamson synthesis

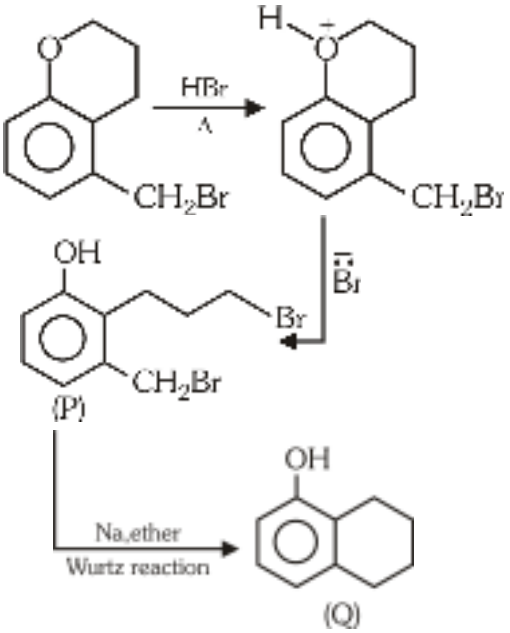
69) Less hindered alkyl halide give S_N2 reaction at fastest rate.

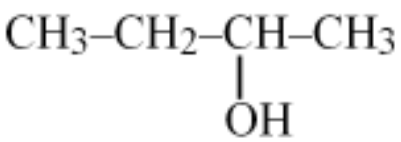
70)

Order of net e^- withdrawing tendency
– I > –Br > –Cl > – F

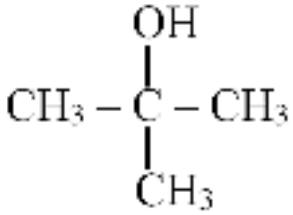
71) 

Here the reaction is as follows



72) 

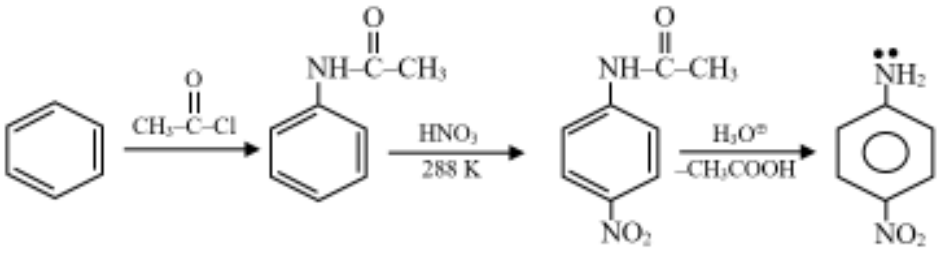
Show CAN test turbidity obtained after 5 min. iodoform test = +ve



Show CAN test turbidity obtained immediately iodoform test negative.

73) Chlorine is an electron withdrawing group but it is ortho, para directing in electrophilic aromatic substitution because Inductive effect of chlorine destabilises the intermediate carbocation formed during the electrophilic substitution, however due to the more pronounced resonance effect, the halogen stabilises the carbocation at ortho and para positions.

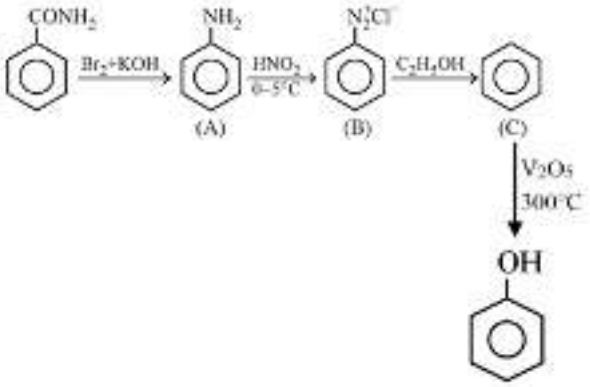
74) **Question Explanation :**
In the given chemical reactions, the major product is :
Concept :
Isocyanide test or carbyl amine reaction.
Solution :



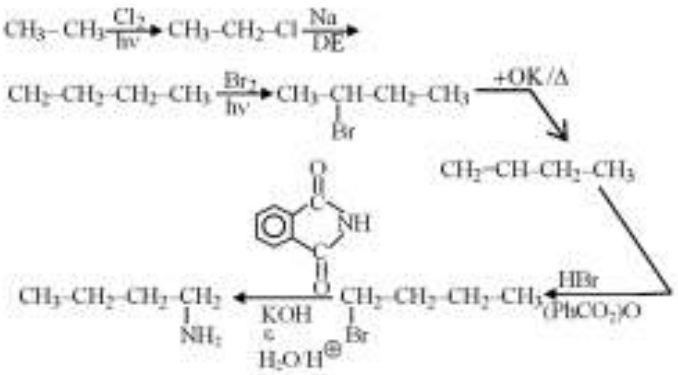
Final Answer : Option (2)

75)

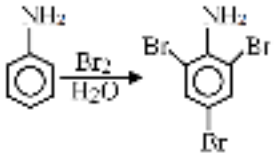
Explain Question: The final product (C) formed in the given reaction is.
Concept:
(i) Hoffmann bromamide reaction.
(ii) Aniline reaction with nitrous acid.
(iii) Reaction of diazonium salt with ethanol.
Solution:



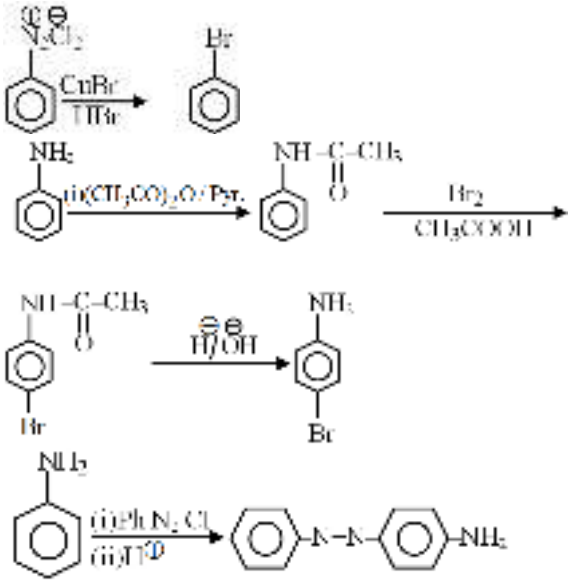
Final answer: option (2)



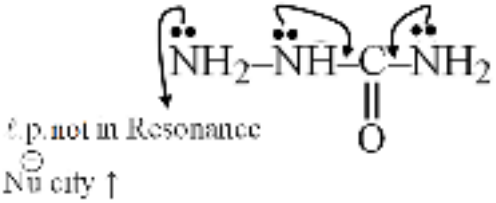
76)



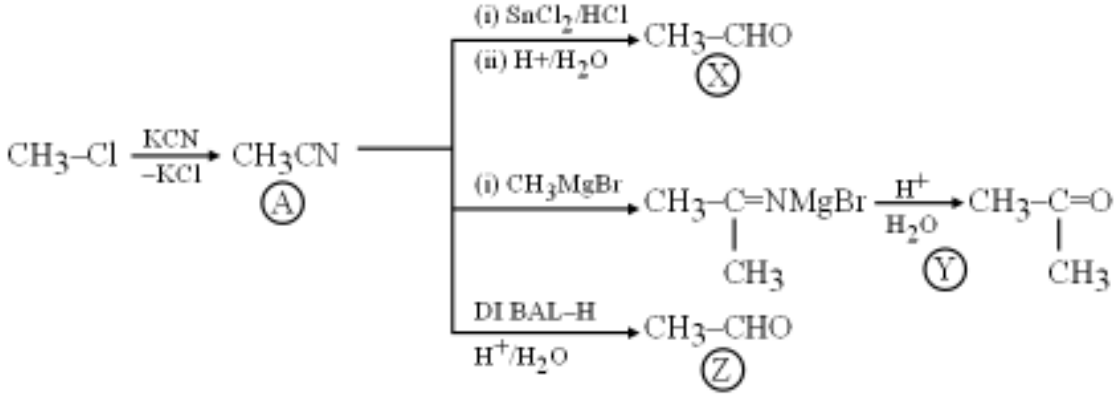
77)



78) If ⊖ .p in resonance than Nu city ↓es



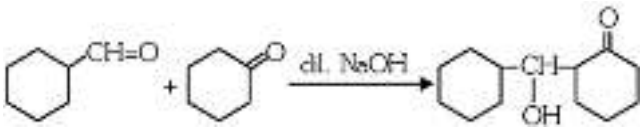
79) **Explain the question :** Identify incorrect statement.
Concept : This question is based on



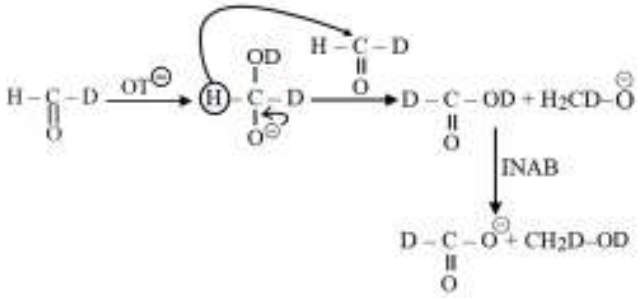
Solution : Tollen's test → given by aldehydes (Y) doesn't give tollens'

⇒ Haloform test → $CH_3-C(=O)-$ containing group gives
⇒ is methyl cyanide
⇒ Rate of NAR : Aldehyde > ketone
(X) (Y)
Final answer : 2

80) **Question Explanation :** Identify Product which can not be formed in given reactions.
Concept : Cross Aldol condensation.
Solution :



Final Answer : Option (3)



81)

82) **Explain the Question :** Evaluate whether the boiling points of aldehydes and ketones are higher than those of hydrocarbons and ethers of similar molecular mass, and whether the reason explains this trend.

Concept : This question is based on physical properties of carbonyl compounds.

Solution :

Assertion : Aldehydes and ketones have higher boiling points than hydrocarbons and ethers of comparable mass.

True : Due to stronger dipole–dipole interactions.

Reason : Weak molecular association arises from dipole–dipole interactions.

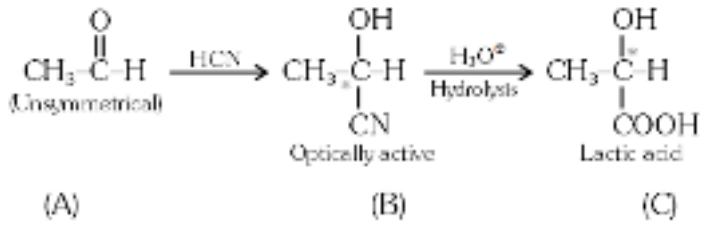
True: Carbonyl groups create partial charges that attract neighbouring molecules.

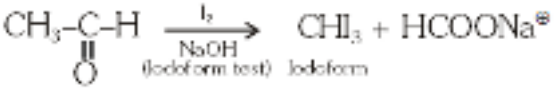
The reason correctly explains the assertion.

Final answer : 1

83)

All aldehyde & methyl aliphatic ketone give white crystal with $NaHSO_3$





84) (A)

85)

Explain Question : The question asks to correctly match the analytical methods in Column-I with the substances or concepts they are associated with in Column-II.

Concept : This question is based on Qualitative and Quantitative Analysis of Organic Compounds, Chromatography.

Solution :

Column-I	**Column-II**	**Principle**	**Match**
(I) Dumas method	(c) Nitrogen gas	Nitrogen is converted to N_2 gas for measurement.	I → c
(II) Kjeldahl's method	(d) Ammonium Sulphate	Nitrogen is converted to Ammonium Sulphate.	II → d
(III) Carius method	(a) $AgNO_3$	Halogens are precipitated as silver halide using $AgNO_3$.	III → a
(IV) Chromatography	(b) Silica gel	Uses a stationary phase like Silica gel for separation.	IV → b

Final Answer : option (1) I→c, II→d, III→a, IV→b

86)

A) Question Explanation: The question asks to check correctness of Assertion and Reason in thin layer chromatography.

B) Concept Used: Thin layer chromatography, adsorption chromatography, silica gel adsorbent.

C) Answer Explanation:

A. Assertion: TLC is adsorption chromatography → True.

B. Reason: Silica gel layer acts as adsorbent → True.

Both Assertion and Reason are true and Reason is the correct explanation of Assertion

D) Final Answer: Option (4)

87)

$1\ NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$
1 mole of H_2SO_4 neutralizes 2 moles of NH_3
moles = Molarity × Volume
moles of $H_2SO_4 = 1M \times \frac{10}{1000} = 0.01$ mole
moles of N = 2 × moles of H_2SO_4
= 2 × 0.01 = 0.02 moles.
mass of N = moles × atomic mass of N
= 0.02 × 14 = 0.28 gm

% of nitrogen = $\frac{\text{mass of N}}{\text{mass of organic compound}}$

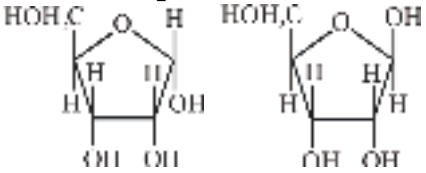
% of N = $\frac{0.28}{0.5} \times 100 = 56\%$

88)

Question Explanation: The question asks which compound is an anomer of the given compound.

Concept Used : This question is based on anomers are stereoisomers that differ only at the anomeric carbon (C1).

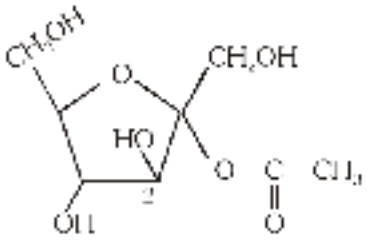
Answer Explanation:



The correct option will be the structure where only the orientation of –OH at C1 differs compared to the given compound, while all other stereocenters remain the same.
Final Answer: Option (2)

89)

Fact

90) 

Reducing sugar in basic medium.

BIOLOGY

91) NCERT Pg. No. # 60

92) NCERT Pg. No. # 81

93) NCERT, Pg. # 72

94) NCERT Pg. # 65

95) NCERT Pg#73,74(E), 81,82(H)

96) NCERT Page-77

97)

NCERT XII Pg. No. # 61
Explain Question : The question asks for the number of possible genotypes and phenotypes for the children of a couple with the blood types AB (I^AI^B) and A (I^AI^O).
Concept : This question is based on Multiple Alleles.
Solution :
I^AI^A: Blood type A
I^AI^O: Blood type A
I^AI^B: Blood type AB
I^BI^O: Blood type B
The possible phenotypes are A, B, and AB. and 4 different genotypes possible for the children.
Final answer : Option (4)

98) NCERT Pg. # 69, 70

99)

NCERT-XII, Pg # 74 (2022 edition) Figure 4.15

100)

NCERT-XII Pg. # 88

101) NCERT-XII, Pg # 85, 86

102) NCERT-XII, Pg # 96

103) NCERT, Pg. # 95

104)

NCERT XII Pg. No. # 92
Question explanation: The question asks to identify the incorrect statement about genes and transcription units.
Concept :
Transcription
Explanation :
1: Incorrect. The promoter and terminator flank the structural gene in a transcription unit, not translation unit.
Final answer : The promoter and terminator flank the structural gene in a translation unit.

105) NCERT (XII)th Pg. # 99,100

106) NCERT Pg. # 94-95

107) NCERT Pg. # 92-93/103

108)

NCERT, Pg. # 83

109) NCERT Pg.#102.

110) NCERT-XII Pg#105(E), 115,116(H)

111) NCERT Pg. # 99

112) NCERT Pg. No. # 93

113) NCERT Pg. # 90

114) NCERT XII Pg. No. # 197-198

115)

Species diversity is more where no. of species are more

116) NCERT XII, Pg. # 198

117) NCERT XII, Pg. # 196, 197, 198

118) NCERT XII, Pg. # 199, 200, 201

119)

NCERT-XII, Pg.# 194 [$N_t = N_o e^{rt}$]
$rt = 0.4 \times 5 = 2$
$Nt = 100 \times e^2$
$e = 2.72$
$e^2 = 7.39$
$Nt = 100 \times 7.39$
$Nt = 739$

120)

NCERT Pg. No. # 196

121)

NCERT XII, Pg # 230

122)

NCERT Pg. No. # 191

123) NCERT XII Pg. # 244 (E), 269 (H)
All given factors

124)

NCERT XII Pg. No. # 206-207

125)

NCERT Pg. No. # 212-213

126)

NCERT XII Pg. No. # 209

127) NCERT Pg. # 245

128)

NCERT XII Pg. No. # 210

129)

NCERT XII Pg. # 182, 183, 185, 187

130) NCERT XII Pg. No. # 229
A-snake, B-Grasshopper, C-Rabbit, D-Deer

131) NCERT, Pg # 224,225

132)

Module Nile perch was introduced not Cichlid.

133)

NCERT Pg. No. 210,225

134)

NCERT XII Pg. No. # 221-222

135)

NCERT XII Pg. No. # 221

136)

NCERT XII, Pg # 179

137)

NCERT XII, Pg # 163

138)

NCERT XII Pg # 174, fig 9.7 (a)

139)

NCERT XII, Pg # 184

140)

NCERT XII, Pg # 171, 172, 174

141)

NCERT XII, Pg # 172

142)

NCERT XII, Pg # 162, 165, 168, 169

143)

NCERT XII, Pg # 165, 166

144)

NCERT XII, Pg # 173, 174

145)

NCERT XII, Pg # 178

146)

NCERT XII, Pg # 179

147)

NCERT XII, Pg # 179, 180, 184

148)

NCERT-XII, Page # 184

149)

NCERT XII, Pg # 182

150)

NCERT-XII, Pg. # 180,183,184

151)

Module

152)

NCERT XII, Pg # 182

153)

NCERT XII, Pg # 178, 180, 184

154)

NCERT XII, Pg # 152

155)

NCERT XII, Pg. # 151

156)

NCERT XII, Pg. # 151

157)

NCERT XII, Pg. # 151, 157

158)

NCERT XII, Pg. # 136

159)

NCERT XII, Pg. # 142

160)

NCERT XII, Pg. # 130, 131, 133

161)

NCERT XII, Pg. # 131

162)

NCERT XII, Pg. # 133

163)

NCERT XII, Pg. # 135

164)

NCERT XII, Pg. # 131

165)

NCERT XII, Pg. # 141

166)

NCERT XII, Pg. # 135

167)

NCERT-XII_Pg. # 136

168) NCERT XII, Pg. # 137, 169-174

169)

NCERT XII, Pg. # 155

170)

NCERT XII, Pg. # 140,141

171)

NCERT XII, Pg. # 139

172)

NCERT XII, Pg. # 129,130,131,132

173)

NCERT XII, Pg. # 131

174)

NCERT XII, Pg. # 133

175) NCERT XII, Pg. # 124

176)

NCERT XII, Pg. # 124

177)

NCERT XII, Pg. # 122

178)

NCERT BASED
Question Explanation: Identifying the correct sequence of plant evolution.

Concept: Plant evolution.

Solution: The correct sequence of plant evolution is: Chlorophyte ancestors → Tracheophyte ancestors → Rhynia-type plants → Psilophyton180.

Final Answer: A. Chlorophyte ancestors → Tracheophyte ancestors → Rhynia type plants → Psilophyton.

179) NCERT BASED

180) NCERT Page No. 116, 120